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Pressing

The press section is the next part of the paper machine after the forming section. The function of pressing is to continue the water removal process which started in the forming section, consolidate the sheet, give texture to the sheet surface, support and transfer the sheet. The fabrics in this section are called press fabrics or press felts. The term “fabric” has a broader meaning; “felt” is a type of fabric which is made of individual fibers only, i.e., no yarn in the fabric structure. Nevertheless, the terms “press fabric” and “press felt” are used as identical terms in the papermaking industry. Although it depends on sheet grade and paper machine, typical sheet consistency at the beginning of the press section is 20% fiber and 80% water and at the end of the press section is 40% fiber and 60% water. At the end of the press section, the sheet is transferred to the dryer section.

During pressing, the sheet is compressed between one or two fabrics and two rolls in the press nip to squeeze water from inside the web and out of the felt fibers. Figure 3.1 shows this process in a plain press nip. Increased compression increases water removal [1–3].

The main functions of a press fabric are water removal from the sheet, sheet support and transportation, providing uniform pressure distribution, and imparting proper surface finish to the sheet. The fabric should provide proper protection for the sheet to resist crushing, shadow marking and groove marking. Other functions include transferring the sheet

from one position to another in case of closed draws, driving the undriven felt rolls and cylinder formers. The amount of water that the felt can absorb and the water flow resistance are affected by the void volume (volume that is not occupied by fibers or yarns) and air and water permeability of the fabric. Low flow resistance and ability to maintain void volume under load are important during operation. Important press fabric properties include strength, adequate void volume, required permeability, low compressibility, batt/base ratio, compaction resistance, abrasion resistance, contaminant resistance, heat and chemical resistance.

3.1 Water Removal Theory

The basis weight of the sheet and its drainage characteristics are critical factors in determining the mechanism for water removal. There are basically two types of nips that define the water removal process.

3.1.1 Flow-Controlled Nips

In these nips, water removal is primarily influenced by water flow resistance in the sheet. In flow-controlled nips, the resistance to fluid flow within the mat of fibers controls the rate at which water can be pressed. These

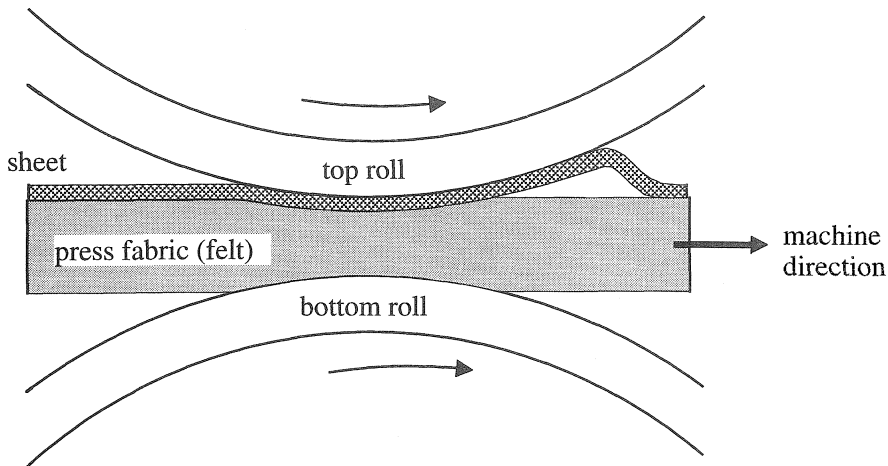


FIGURE 3.1. Schematic of simple press nip.

nips are characterized by high water loads, heavyweight sheets, slow draining and low freeness stocks.

Major symptoms of problems in flow-controlled nips are crushing and hydraulic flow mark. Water removal is aided by:

- soft roll covers
- large diameter press rolls
- double felting
- high sheet temperatures

Design Considerations

- (1) Low flow resistance: In flow-controlled nips, it is very important that flow resistance be minimized. Typically, coarser batt deniers and higher permeability fabrics are used to facilitate water removal without crush or hydraulic mark.
- (2) High void volume: Because of the higher water loads, high void volume structures are usually required. Generally multilayer or laminated fabrics are used to handle the water flow in the nip.
- (3) Flow-controlled nips, where the sheet structure restricts water flow, may occur with lightweight, low freeness sheets, such as carbonizing, condenser and glassine: In this case, coarser batt denier would not be applicable, and void volume requirements would be less.

3.1.2 Pressure-Controlled Nips

In pressure-controlled nips, the mechanical resistance to compression within the sheet controls the rate of water removal. These nips are characterized by low water loads, lightweight sheets, fast draining and high freeness stocks.

Major symptoms of problems in pressure-controlled nips are vibration, poor consistency and physical impression mark. Water removal is aided by:

- hard roll covers
- small diameter press rolls
- pressing uniformity
- high sheet temperatures

Design Considerations

- (1) Base pressure uniformity: Higher sheet contact during pressing is very important. High contact base designs are preferred.
- (2) Mass uniformity: Because of the higher nip intensity and peak nip pressures, press bounce and vibration are potential problems.
- (3) Batt stratification: Finer deniers are used to improve the pressure uniformity of the fabric. These are usually needled over coarser fibers for permeability control and resistance to filling.

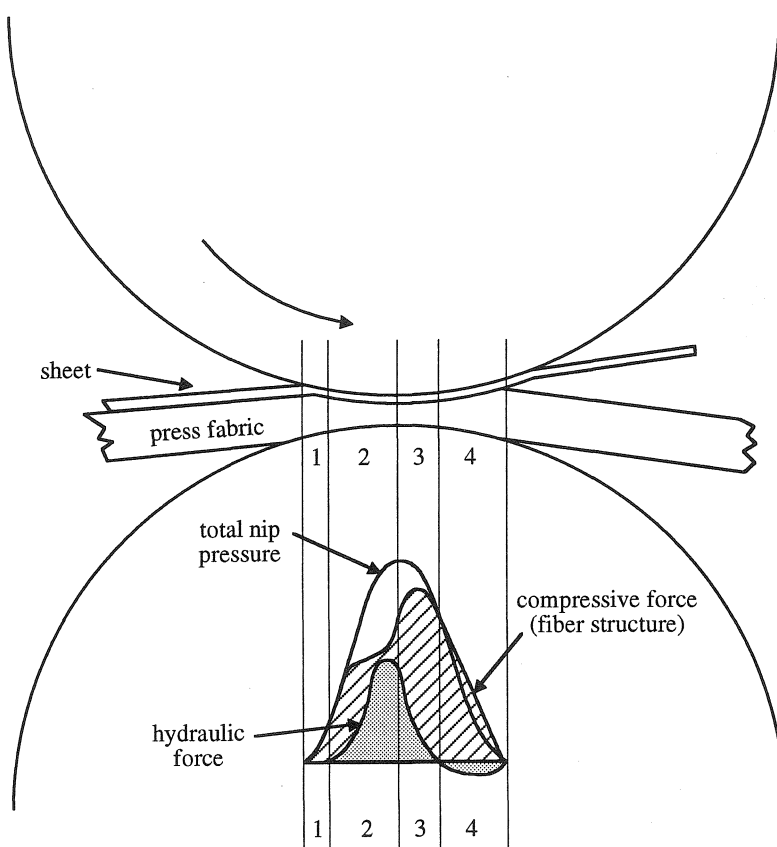


FIGURE 3.2. Transversal flow nip.

An understanding of the fundamentals of transversal flow pressing is necessary to optimize press and press fabric design. Figure 3.2 shows a typical transversal flow press nip in which press operation has been broken down into different phases based on mechanisms involved in water transfer. The nip is defined by two solid rolls with paper and fabric passing through the nip. Both the press fabric and paper are unsaturated entering the nip.

Phase 1 starts at the entrance of the nip where pressure develops in the sheet structure and continues until the sheet is saturated. There is no hydraulic pressure generated in this phase.

Phase 2 begins when the sheet becomes saturated and hydraulic pressure develops. Water is pressed from the sheet into the press fabric. If the press fabric becomes saturated,

hydraulic pressure causes flow of water into the voids of the roll (grooves, holes, etc.). This phase extends to the mid-nip or the point of maximum pressure.

Phase 3 extends from the point of maximum total pressure to the point of maximum dryness. Maximum dryness occurs at maximum structure pressure and where the hydraulic pressure in the paper reaches zero.

Phase 4 begins where the paper and press fabric start to expand resulting in negative hydraulic pressure. Both the fabric and paper become unsaturated and rewetting occurs due to capillary forces and pressure differences between the press fabric and paper. During the expansion phase, the sheet and fabric compete for the boundary water.

The press fabric is a necessary component of this water removal mechanism. The press

fabric provides a porous structure to which the water can flow from the paper in the incoming nip and it should retain this water in the expansion phase of the nip. The ideal fabric should provide perfectly uniform pressure distribution, lowest possible flow resistance, and smallest rewet in the outgoing nip.

An analysis of nip conditions using Walstrom's model [4] can help identify the terms of our compromise. The following equation represents a model for outgoing moisture ratio for a nip.

$$MR_{\text{out}} = k - f_1 - f_2 - f_3 - P - RW - R_1 - R_2 \quad (3.1)$$

where

MR_{out} = outgoing moisture ratio

k = moisture ratio of maximum paper dryness at zero flow resistance and uniform pressure distribution determined by compression characteristics of the paper

f_1 = increase in moisture ratio due to flow resistance in the fiber wall

f_2 = increase in moisture ratio due to flow resistance in the paper structure

f_3 = increase in moisture ratio due to flow resistance in the press fabric

P = increase in moisture ratio due to nonuniform pressure application

RW = increase in moisture ratio due to rewetting, redistribution of water between paper and fabric

R_1 = increase in moisture ratio due to rewetting of the press fabric from the pressure structure

R_2 = increase in moisture ratio due to rewetting after the nip

A closer look at this model suggests how press fabric properties influence water removal.

f_3 , the effect of fabric flow resistance, is dependent on speed, capillary structure, incoming moisture, moisture change, temperature, and basis weight. It is worth noting that open press fabric structures, to reduce flow

resistance, are in direct conflict with the property of uniform pressure distribution.

P , the effect of uniform pressure distribution, is likely determined primarily by the press fabric. The fabric should bridge the grooves, suction holes, etc., to exert a uniform pressure to the paper.

RW , R_1 and R_2 concern rewet and are dependent on press nip conditions, paper and fabric structure and their dryness.

The relative importance of each factor and the direction of fabric design compromise are determined by the following:

- water load
- nip pressure and width
- speed or nip dwell time
- sheet properties; freeness, furnish and weight
- temperature

At low basis weights, free sheets, and low speeds, flow resistance in the paper or fabric (f_2 , f_3) is relatively negligible; P and rewetting dominate. Smooth, dense fabrics are preferred. At the other extreme—high basis weights, low freeness, and high speeds—flow resistance is highly important. Double felting and/or high-void volume fabrics can be of benefit. Pressure distribution and rewetting continue to be important.

In between the two extremes, all press variables need to be considered. Also it should be noted that the relative importance of each of these variables changes with water load. First presses with high water loads tend to behave more as flow-controlled nips, even on lightweights. Last presses have lower water loads and move toward pressure-controlled conditions. Table 3.1 shows the relative importance of fabric characteristics for pressure- and flow-controlled nips.

Figure 3.3 shows the distribution of hydraulic and compressive forces at the first, second and third nip in a press section. Hydraulic force is greater in early presses; compressive forces are higher in later presses. Table 3.2 lists typical sheet consistencies for various grades.

TABLE 3.1. Relative Importance of Fabric Characteristics for Pressure- and Flow-Controlled Nips.

Fabric Characteristics	Pressure-Controlled Nip	Flow-Controlled Nip
Caliper	↓	↑
Permeability	↔	↑
Void volume	↔	↑
Pressure uniformity	↑	↗

3.2 Press Fabrics (Felts)

3.2.1 History

Until the 1930s papermakers' felts had improved very little over the woven or pressed woolen fabrics observed in China by Marco Polo. In this pre-World War II era chemical treatments were developed to alter the woolen molecule. This improved the ability of wool to resist chemical, mechanical and bacterial degradation. In 1946 the introduction of staple man-made fibers contributed greatly to the strength and durability of wet felts. Early attempts to full or felt 100% synthetics resulted

in fabrics that were too open or porous and unstable.

Since about 1944, the asbestos cement industry had been using fabrics made by a process known as needling. This was the process of laying a carded web of textile fibers on a woven base and needle-punching the web fibers into the base with barbed needles. It was considered that higher synthetic, unfilled fabrics might be made by this method to not only achieve desirable properties of woven, full felts, but also to make improvements in terms of life, finish and drainage. Needled fabrics were then produced in the synthetic ranges from 35% to 49%. In the late sixties, 75% to

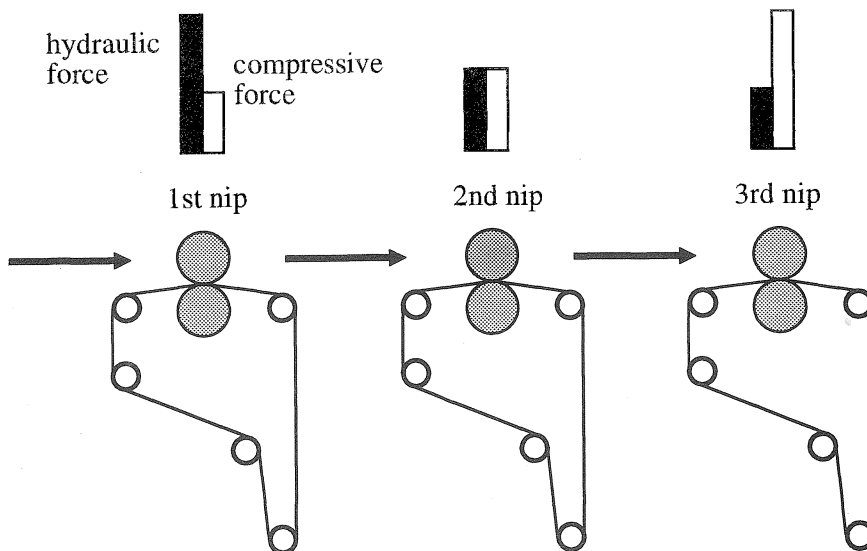


FIGURE 3.3. Distribution of hydraulic and compressive forces at different nips.

TABLE 3.2. Typical Press Dryness Number [5].

	Load		Exit Sheet Consistency, %
	kN/m	pli	
Newsprint			
First press	60	300	33
Second press	80	450	38
Third press	95	550	43
Bond			
First press	60	350	37
Second press	80	450	41
Third press	95	550	43
Corrugating medium			
First press	70	400	37
Second press	105	600	41
Third press	315	1800	46
Linerboard single felted			
First press	70	400	35
Second press	105	600	39
Third press	175	1000	42
Linerboard double felted			
First press	105	60	38
Second press	210	1200	43
Third press	350	2000	46

100% synthetic needled fabrics appeared and provided what some thought was the optimum in felt construction.

The introduction of the fabric press (inner fabric), whereby an incompressible fabric of monofilament or highly twisted multifilament synthetic is run between the felt and the press nip, resumed the evolution of the wet felt. The principle of the fabric press was built into the needled felt using a base fabric in mesh construction which provided significantly greater void volume within the total fabric structure. This improved the water handling capacity of the felt, allowing increased press loading. The fabric was easier to clean, and its tremendous strength and comparative rigidity provide remarkable improvements in felt life. The batt-on-mesh felt was, therefore, developed and felt-making took another step forward to the benefit of the papermaking industry.

3.2.2 Press Fabric Functions

Once the sheet has left the headbox, the general requirement of the paper machine is to increase the fiber consistency of the sheet from 0.2–1.5% to 92–96%. After the forming fabric, water removal costs far less in the press section than the dryer section. The value of efficient press fabric performance, therefore, can not be overemphasized.

Water removal, however, is not the only function of the press fabric. Its basic performance criteria includes other functions as well. In general, the press fabric must:

- (1) Accept the water that is expressed from the sheet in the press nip.
- (2) Provide the proper protection for the sheet to:
 - resist crush
 - resist shadow mark

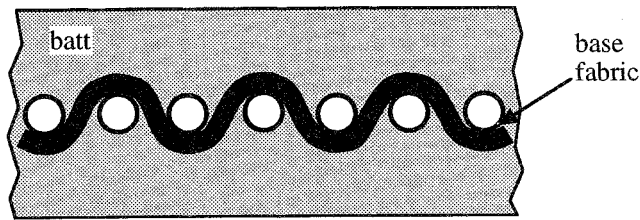


FIGURE 3.4. Schematic of a typical press fabric structure.

- resist groove mark
 - resist base fabric mark
- (3) Present the proper surface to the sheet so that the necessary degree of smoothness or finish requirement is imparted to the grade of paper to be manufactured.
 - (4) In case of closed draws, transfer the sheet from one position to another
 - (5) Provide desirable durability in terms of strength, resistance to mechanical abrasion, chemical degradation and fill-up.

3.2.3 Construction of Press Fabrics

Press fabrics, in general, consist of two components as shown in Figure 3.4: base fabric and batt.

The Base Fabric

Today, most press fabrics are made of 100% synthetics, primarily polyamide (nylon) polymers. Base fabrics are usually woven or constructed with cabled monofilament, plied multifilaments, spun yarns and/or single monofilaments (Figure 3.5).

Each yarn has properties that influence the operational characteristics of the press fabric. They are designed into the weave of the base

fabric to affect sheet quality, water removal performance, runnability and ease of installation. Primary characteristics of the more commonly used yarns are given in Table 3.3.

The base fabric may be a single layer or multilayer mesh. The fabrics can be woven endless or woven flat and joined with a seam (Figure 2.3, Chapter 2). The weave of the base fabric is engineered to manipulate pressure uniformity, flow resistance, void volume, and compression properties. The basic classifications of press fabrics are conventional (endless) designs, stratified (laminated) designs and seam fabrics. Figure 3.6 shows the main types of base fabrics used in press fabrics.

Seam fabrics have shown a substantial increase in usage since their introduction in the late 1980s. The primary factors for the increase in use of seam products are safety, reduced installation time and, in some cases, improved press fabric performance. There are basically two types of seam products as shown in Figure 3.7.

Pin seam fabrics are woven flat with loops woven into the fabric in a separate process. These fabrics are characterized by crimp in the machine direction yarns. Woven loop seam fabrics are woven endless with loops formed during the weaving process. Like conventional endless fabrics, the woven loop design

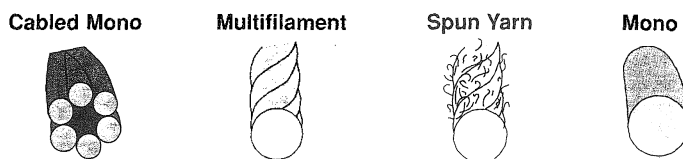


FIGURE 3.5. Yarn types used in manufacture of base fabrics.

TABLE 3.3. Characteristics of Multifilament and Monofilament Yarns Used in Press Fabrics.

Multifilament	Monofilament
Very durable	Durable
Supple	Stiffer than multifilament
Compressible	Resists compression
Higher elongation	Less stretch than multifilament
Low resistance to chemical attack	Better resistance to chemical attack

has crimp in the cross machine direction. Both products can also be made with laminated fabrics on the surface.

The Batt

The process in which the batt is locked to the base fabric is called needling. The batt is carded into a uniform web first and then is applied in a series of layers onto the base fabric. The web and base fabric are fed through a zone where several thousand reversed barbed needles are needle-punched into the composite to lock the web to the base fabric. The batt is typically spliced at the start and stop of web application. Some processes can apply the web in a spiral method that eliminates the cross machine direction oriented splice. These processes are generally

applied on fabrics for critical press bounce and vibration prone positions.

The needling process can be engineered to affect the density, surface properties and permeability of the press fabric. The batt fibers that are used in the manufacture of press fabrics are purchased in standard sizes. The most commonly used unit to indicate the density of the fiber or yarn is the denier. Denier is actually a weight measure but it has become synonymous as an indicator of fiber fineness (Section 2.2.3). As long as the specific gravity of the polymer is approximately the same, denier can be used to compare fiber diameter. The relative sizes of the most common fibers used in press fabrics are shown in Figure 3.8. As can be seen from the figure, a 30 denier fiber is not twice as large as a 15 denier, and a 6 denier fiber is not twice as large as a 3

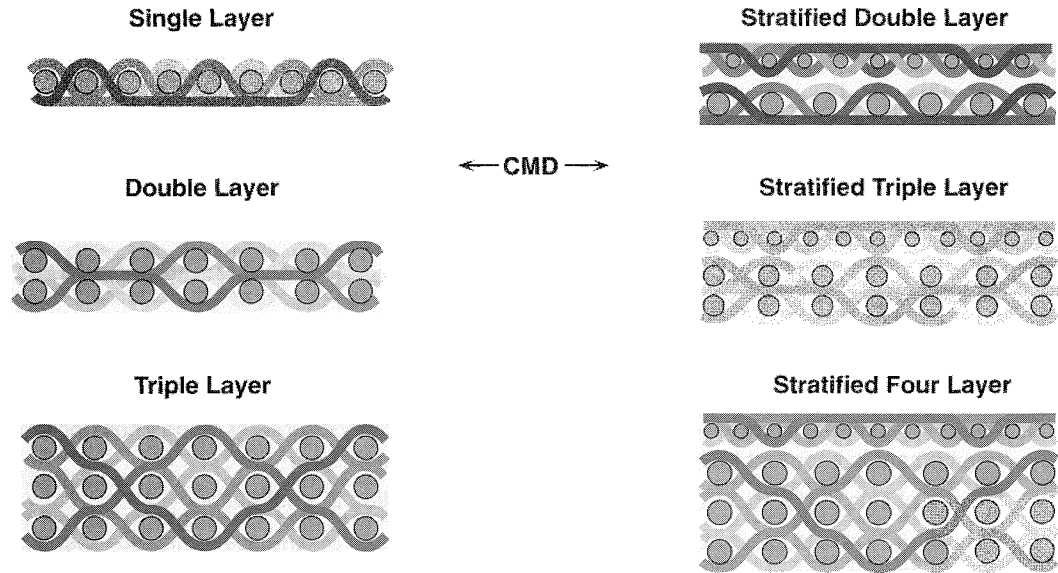


FIGURE 3.6. Major types of base structures used in press fabrics.

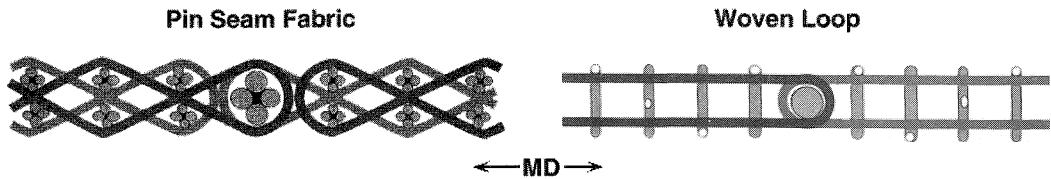


FIGURE 3.7. Seam fabrics.

denier fiber. The relation between denier and diameter is explained in Section 2.2.3 for some common paper machine clothing materials.

Many factors influence the selection process of the fiber denier. These include:

- water handling requirements of the fabric
- break-in time
- available uhle box vacuum
- sheet control issues such as drop offs, blowing and sheet stealing
- fiber shedding propensity
- filling propensity
- bleed through propensity

Fibers with different deniers can be blended or applied in stratified layers for desired performance attributes. Figure 3.9 shows the most common means of batt application.

3.2.4 Press Fabric Designs

Conventional woven felts, which were woven with spun yarns became obsolete. Other major press fabric designs that have been used are explained below. Of these, the first two designs also have become obsolete.

Batt-on-Base Felt

A woven fabric, usually of spun yarns, is used as the base fabric on which fiber batt is needle-punched. Batt-on-base felts were made of either a combination of wool and synthetic fibers or 100% synthetic fibers (mostly nylon), which became the standard. The base fabric, which can have synthetic content between 50–100%, can be woven as endless belt or can be woven flat and joined later. The batt web has synthetic content between 20 and 100%. After washing and chemical treatment, the felt is stretched on the dryer to the operating dimensions. Singeing is usually done to prepare the surface and remove the loose fibers. Finally, the felt is dried at an overstretch length for easy installation. Figure 3.10 shows the random nature and structure of the top surface of a needled felt.

Batt-on-base felt designs have become obsolete and are not used anymore.

Baseless Felt

In this type of structure, no base fabric is used; felt is made by needle-punching only. The absence of base yarns reduces marking

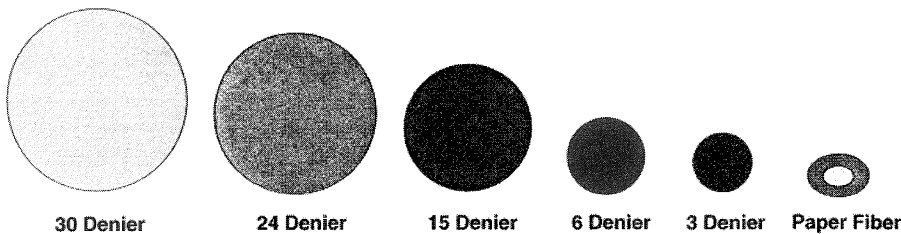
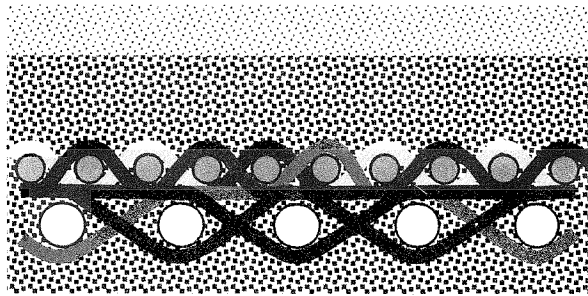
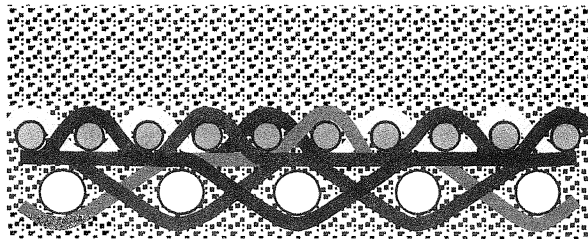


FIGURE 3.8. Relative sizes of the most common fibers used in press fabrics.

Stratified Needling
Fine Denier Over Coarse



Blended Coarse And Fine Deniers



Single Denier Blend

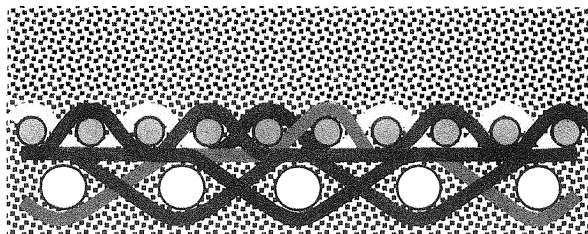


FIGURE 3.9. Most common ways of batt application.

and provides a uniform pressure distribution in the press nip. Nonwoven felts are suitable for suction, grooved and shrink sleeve presses, to provide a smooth sheet finish for fine paper and board machines. This type of design is outdated and not used anymore.

Batt-on-Mesh Felt

Needle-punched batt-on-mesh felt structure is similar to batt-on-base felt structure. Batt-on-mesh fabrics are generally made of

100% synthetic materials. The base structure may be a single or multilayer fabric woven with 100% monofilament, a combination of mono- and multifilament, all high twisted multifilament, all high twisted multifilament treated with resin for rigidity, or any combination of these yarns. Since these yarns are stiffer than spun yarns, batt-on-mesh fabrics have higher resistance to compression and compaction than batt-on-base felts. The base mesh is woven endless and the batt is applied on needle-punching machines. Figure 3.11

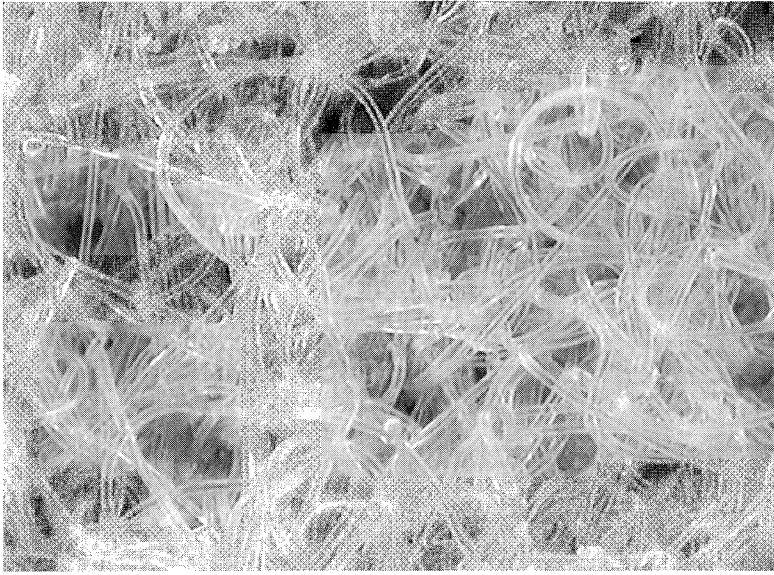


FIGURE 3.10. Random nature of the top surface of a needleless press fabric.

shows a batt-on-mesh felt with single layer monofilament base fabric. Installation of batt-on-mesh fabrics may be more difficult than batt-on-base fabrics due to stiffer base structure.

Multifilament batt-on-mesh fabrics are suitable for suction, grooved and shrink sleeve presses. Resin treated multifilament fabrics are applicable on first and second suction presses for coarse paper grades such as kraft. Monofilament or mono- and multifilament combination base structural felts have excellent stability, resistance to wrinkles, and provide good finish for fine paper applications on suction pick-up positions, straight suction presses, and grooved or shrink sleeve presses.

Felts with No-Crimp Base-Fabric (Knuckle-Free)

In this type of design, the base fabric is not a woven fabric but is formed by two or more, completely separate uncrimped yarn layers. Figure 3.12 shows a two layer no-crimp felt structure. It is reported that the no-crimp base fabric structure provides improved sheet dewatering, good resistance to compaction and

dimensional changes, resistance to filling and improved microuniformity of pressing. Eliminating knuckles in the base structure reduces the possibility of paper fines and dirt being trapped at the yarn crossover points. Moreover, it reduces sheet marking. Other advantages of knuckle-free felts include improved abrasion resistance, easy cleaning and smoother surface. However, due to lack of interlacing of yarns, void volume is drastically reduced and fabric stability may suffer.

Laminated (Stratified) Press Felts

In this relatively new press fabric structure, two or more base fabric structures are included in the fabric. The top base fabric is usually a single layer. The bottom base fabric can be a one, two or three layer integrally woven fabric (Figure 3.13). Laminated fabrics allow a wider range of base fabric pressure uniformity and low batt-to-base ratio which is critical for open and clean operation. As the machine speeds increase, the nip resistance time decreases and better surface contact between the sheet and press fabric becomes a requirement.

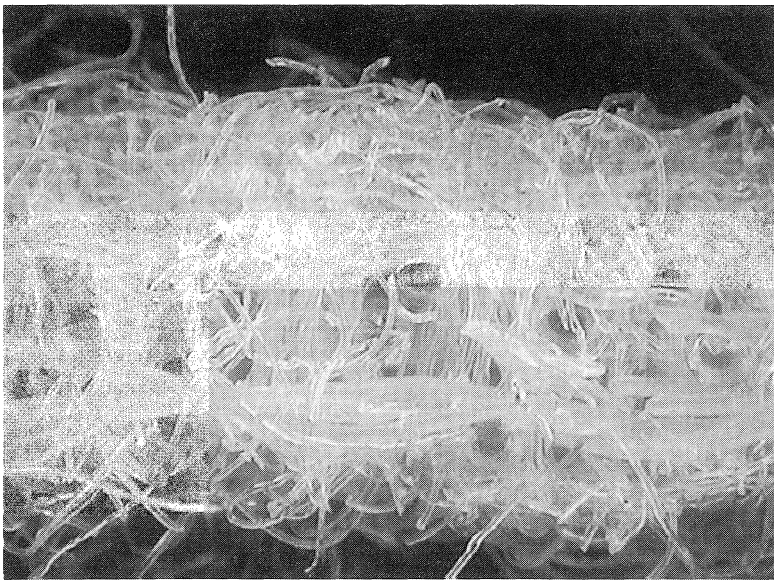
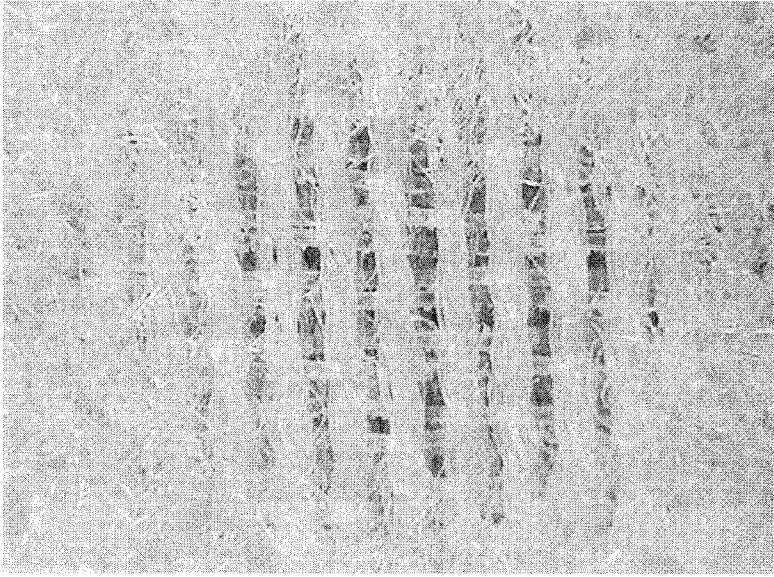


FIGURE 3.11. Batt-on-mesh press fabric structure: paper surface with exposed base fabric and cross-sectional view along the MD.

Continuous research is being done to develop new and improved press fabric designs. The latest developments include utilization of round, hollow yarns in cross machine direction. Hollow yarns flatten under pressure and immediately rebound after exiting the nip as

shown in Figure 3.14. Flexing of yarns provides better sheet contact which in turn improves water removal and smoothness. Machine runnability and printability on the sheet are also improved. Another development is 100% monofilament (no batt) press fabrics.

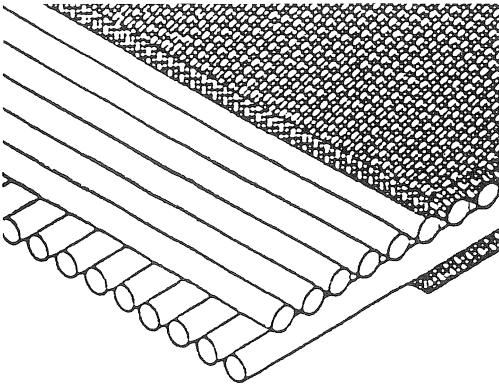


FIGURE 3.12. Two-layer no-crimp base press fabric [6].

Table 3.4 shows the base fabric weight, total fabric weight and air permeability ranges of typical single and multilayer fabrics.

3.2.5 Manufacturing of Press Fabrics

Due to batting, manufacture of press fabrics is different than forming or drying fabrics. Figure 3.15 shows the major manufacturing steps for press fabrics.

Batt fibers are blended, carded and usually preneedled first. Base fabrics are woven (except no-crimp base fabrics) similar to forming fabrics. They can be woven as a flat or endless belt as shown in Figure 2.3 (Chapter 2). The warp (MD) direction of an endless woven belt becomes the CD direction on the paper machine.

Flat woven fabrics have to be joined or pin-seamed to make an endless construction.

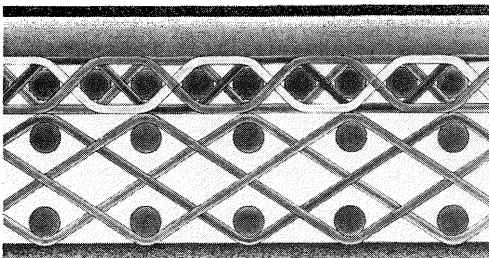
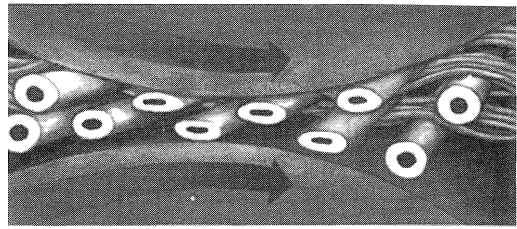
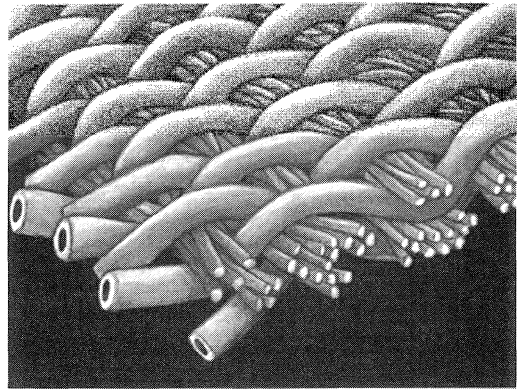


FIGURE 3.13. Cross section of a laminated press fabric.



Process through the nip



base fabric

FIGURE 3.14. Fabric with hollow yarns in CD.

Today, approximately 25–30% of the press fabrics are seamed. Usually pin seaming is used which makes installation of the fabric on paper machine a lot easier. Figure 3.16 shows a press fabric seam. Monofilament warp yarn ends are looped and woven back into the fabric at the two ends of the fabric. A monofilament pin passes through the loops connecting the two ends together after the installation of the fabric on the paper machine.

Endless fabrics are woven in a spiral with the filling yarn forming the length of the fabric. Fabric length will be twice the reed width. Woven loop seam fabrics are woven similar to an endless fabric; however, loops are formed at the ends of the fabric (Figure 3.17). Some looms for endless weaving are as wide as 31 meters (Figure 3.18).

After base fabric manufacturing, carded fiber web is needle-punched into the base struc-

TABLE 3.4. Typical Property Ranges of Single and Multilayer Press Fabrics.

Structure	Base Fabric Weight (oz/sq ft)	Total Fabric Weight (oz/sq ft)	Air Permeability (cfm)
Single layer	0.9–1.8	2.3–4.9	8–160
Two layer	1.9–2.7	3.7–5.8	12–140
Three layer	2.2–2.7	4.0–6.0	15–120
Laminated two layer	2.5–2.7	4.6–5.2	12–90
Laminated three layer	2.2–2.7	4.9–7.0	15–130
Laminated four layer	3.4–3.8	5.2–7.0+	15–130+

ture to form the batt. Batt laying can be applied from preneedled web (Figure 3.19) or as part of a direct lay batting system (Figure 3.20). Figure 3.21 shows the schematic of needling process and a barbed needle used for this purpose. The needle plate moves up and down into the batt and base fabric several hundred times a minute. In the process, the barbs on

the needle grab and penetrate the batt fibers into the base structure. In modern press fabrics, different length and diameter fibers can be used to obtain stratified batt layers, the coarsest fibers typically being next to the base fabric and the finest fibers being on the surface. Special purpose fibers such as fusible fibers can be used to improve bonding.

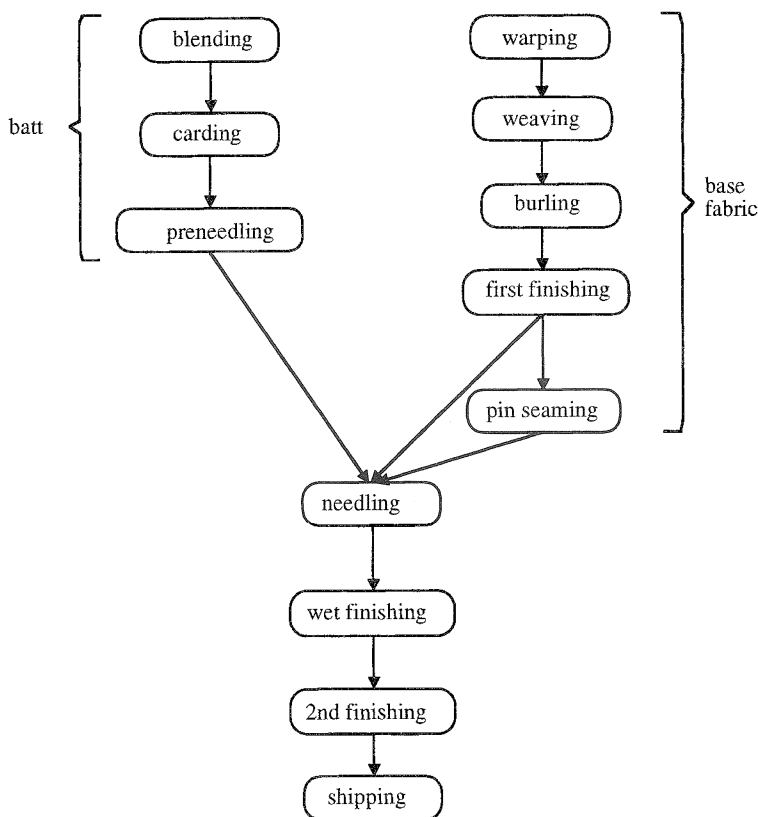


FIGURE 3.15. Manufacturing steps of press fabrics.

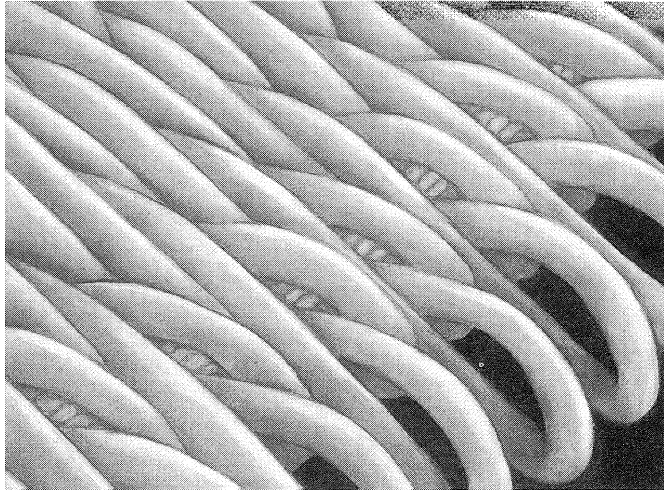


FIGURE 3.16. Pin seam press fabric.

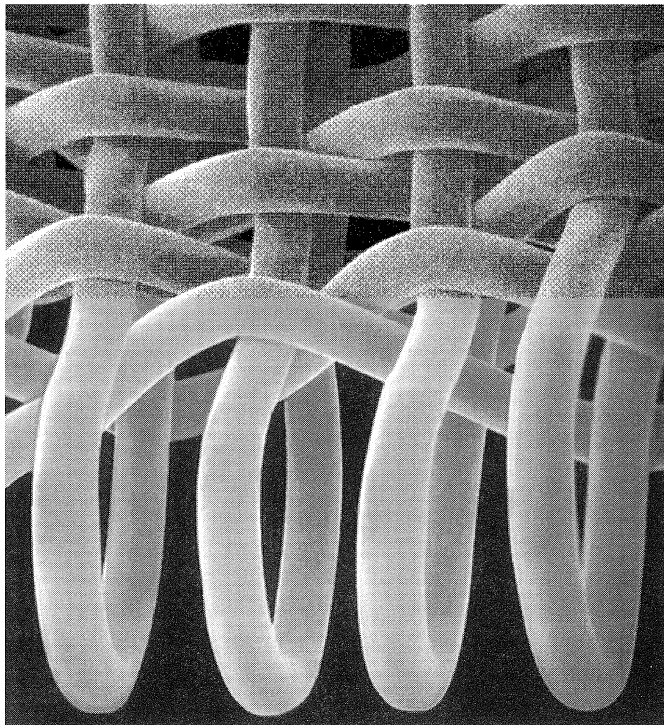


FIGURE 3.17. Woven loop seam fabric.

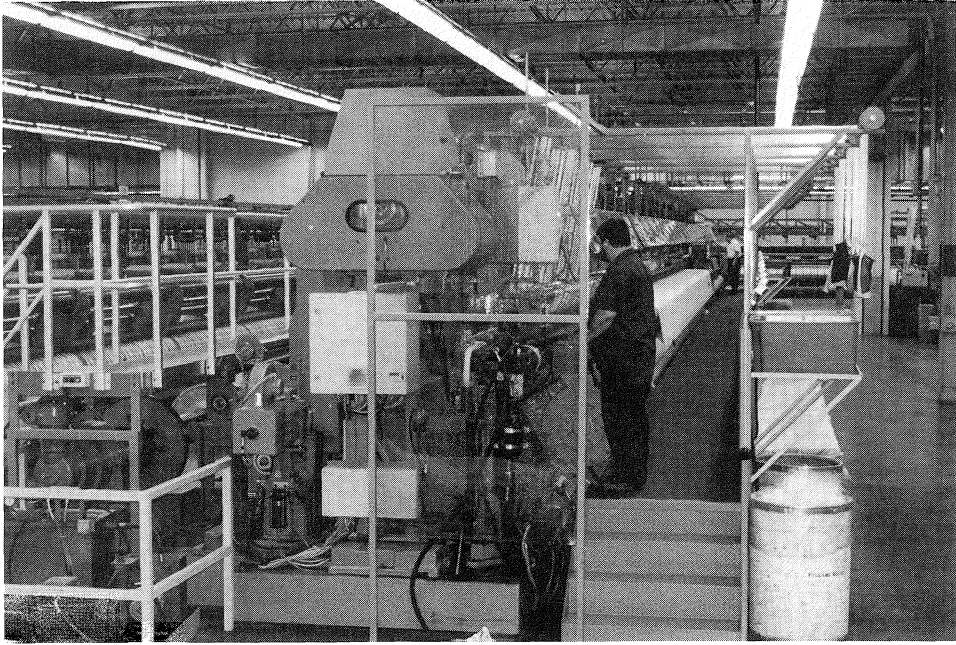


FIGURE 3.18. Weaving machine for endless base fabric manufacturing (machine width = 31 meters).

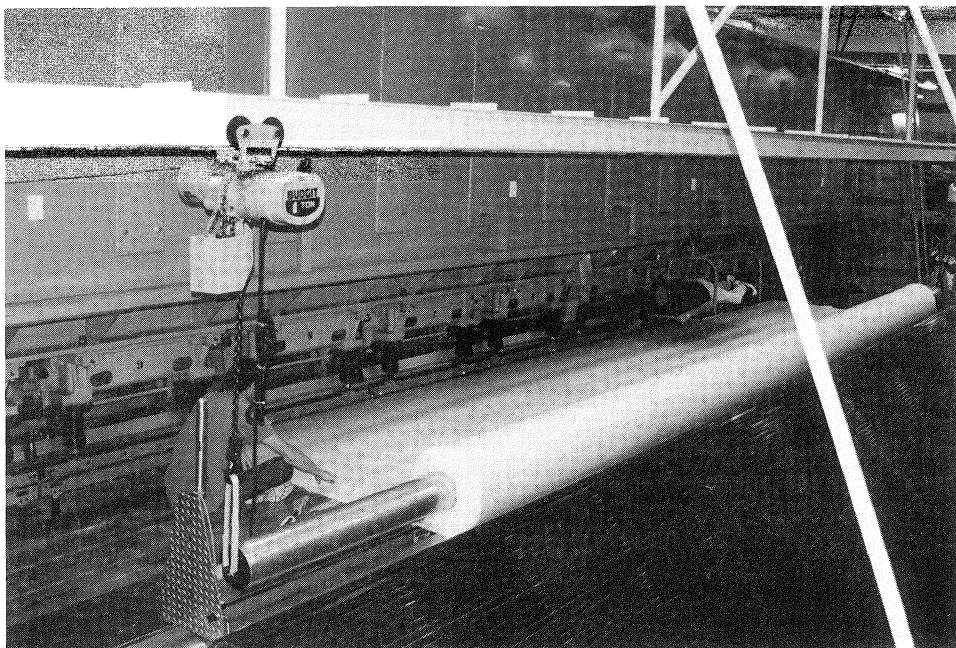


FIGURE 3.19. Preenedding.

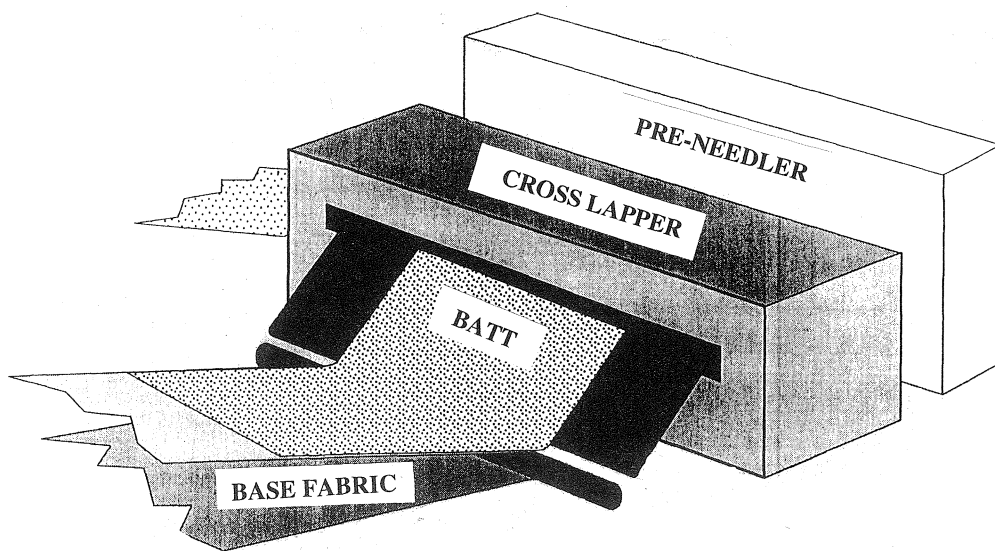


FIGURE 3.20. Direct-lay batting.

barbed needle

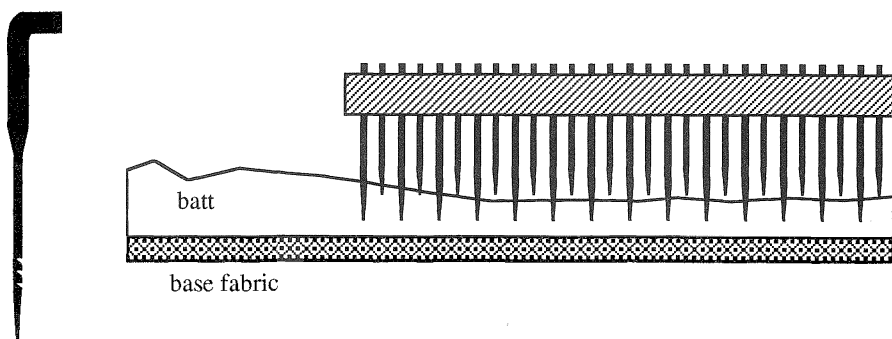


FIGURE 3.21. Schematic of needling.

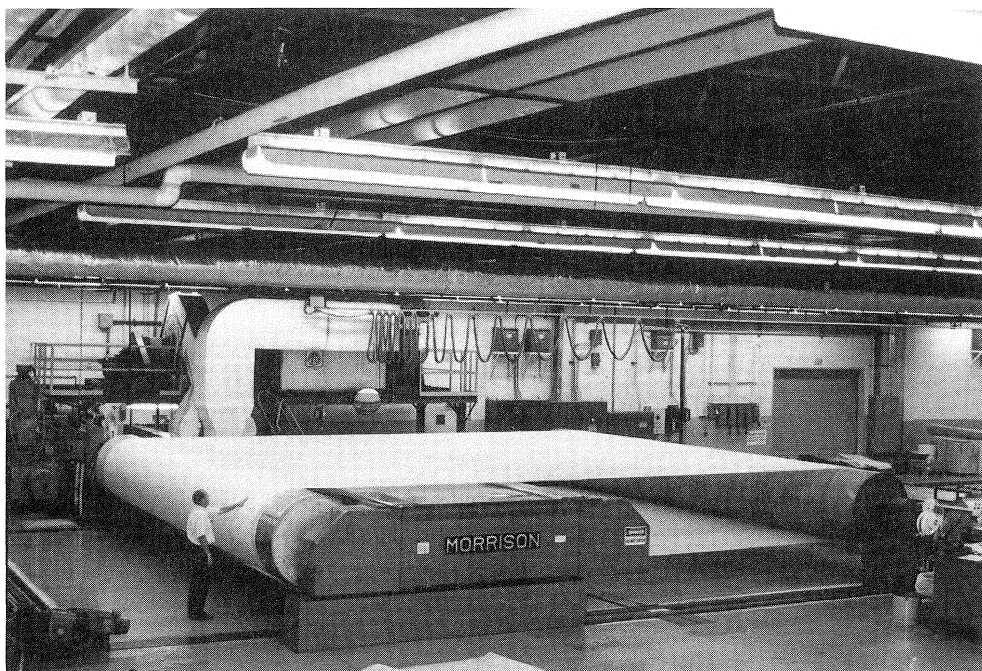


FIGURE 3.22. Press fabric heatsetting.

The remaining major manufacturing steps are heatsetting for stabilization (Figure 3.22), washing (Figure 3.23), treatment and singeing. Treatments can refer to a chemical additive, a special process, or a specialty fiber that is designed to improve the properties of the press fabric. Most treatments are either colorless or reddish-brown in appearance. The most common uses for treatments are:

- improved flexibility
- improved batt adhesion
- shed resistance
- resistance to chemical degradation
- resistance to mechanical wear
- improved compaction resistance
- inhibit contamination
- inhibit wad burns

Singeing is done to remove loose fibers and properly prepare the surface. Press fabrics are sealed in packages prior to boxing in order to protect the fabric against moisture and dirt. The fabric remains in this package until ready for installation. Certain types of press fabrics are rolled and put into a paper tube to prevent possible creasing of the fabric.

3.2.6 Properties of Press Fabrics

Based on their structures (number and dimensions of base yarns, size of batt fibers, the smoothness of the surface finish), press fabrics are classified as superfine, fine, medium and coarse. The important properties of press fabrics are as follows.

Fabric Mass and Thickness

Fabric “mass” is defined as the fabric weight per unit area (oz/sq ft or g/m^2). Press fabric should have a uniform mass distribution across the width. Uneven mass or thickness (caliper) may cause press bounce or vibration. Thickness is important for felt wear and compaction characteristics which influence void volume and drainage of the fabric. The rate of thickness change during the operation depends on the fabric design. The thickness decreases quite rapidly during the early days of the operation. After a certain thickness loss, the fabric has to be removed from the machine. Thicker fabrics are also used to increase nip width and residence time.

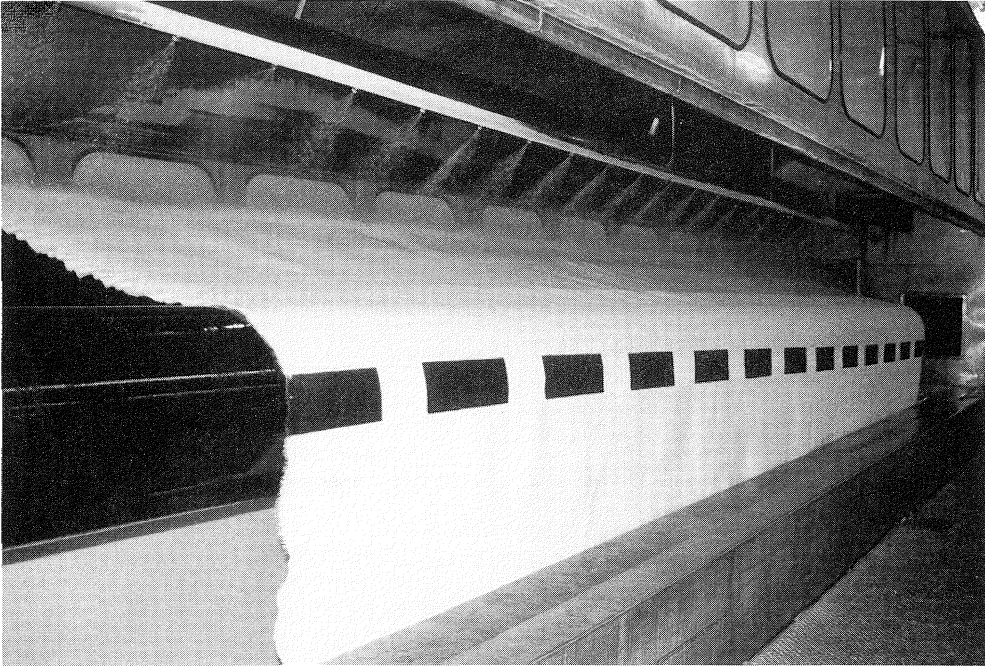


FIGURE 3.23. Press fabric washing.

Batt-to-Base Ratio

Batt-to-base ratio is defined as the mass of the batt fibers divided by mass of the base fabric. As this ratio increases, compactness of the felt also increases, which is not very desirable for dewatering capability. However, a small batt to base ratio may cause marking on the sheet due to base yarns.

Air Permeability

Similar to forming fabrics, it is measured as the air volume (cubic feet) passing through of per unit area (square feet) of fabric per minute (cfm) at 0.5 inch water column.

Void Volume

Void volume is the volume that is not occupied by the yarns and fibers in the fabric. It is an indication of the amount of water that the felt can absorb. Void volumes are measured at various loads to predict the fabric's performance on different types of presses.

Flow Resistance

Properly designed fabrics should provide adequate drainage and filtration of water and therefore should have minimum resistance to flow. Resistance to water flow in the fabric is measured in three directions using water permeability testers: machine direction, cross machine direction and vertical direction. Low flow resistance indicates higher water acceptance capacity of the fabric.

Compressibility and Resiliency

Compressibility is a measure of compactness under load and resiliency is a measure of rebounding capability of the fabric after compaction. Compaction resistance is a desired property in press fabrics. A compaction resistant fabric can be either incompressible or compressible but resilient.

Uniformity of Pressure Distribution

Pressure from the rolls should be transferred to the sheet uniformly by the felt in

order to prevent uneven sheet moisture profiles. Press fabrics with fine base structure and fine batt fibers provide more uniform pressure distribution. Too coarse base yarns, improper weave design or insufficient batt may also cause sheet marking.

Fabric dimensional stability, abrasion resistance, contaminant and chemical degradation resistance are important requirements of modern press fabrics. The fabric should be dimensionally stable throughout its operation. Some fabrics may get narrower after running some time due to heat from furnish and steam boxes. Smooth sheet transfer also requires dimensional stability. Abrasive inorganics in alkaline paper making and paper making chemical additives for mechanical conditioning have increased the importance of abrasion resistance in press fabrics. Increased use of secondary fibers has made contaminant resistance a necessity in today's press fabrics. Paper making chemical additives such as bleaching, oxidizing and cleaning agents require chemical degradation resistance.

3.3 Applications of Press Fabrics

3.3.1 *Installation and Operating Procedures*

Before Installation

- Clean press and other machine components thoroughly.
- Check all machine components such as uhle boxes and roll faces for rough surfaces.
- Ensure that stretch roll and hand guide are perpendicular to the machine.
- Check automatic guide for free movement and guide palm for cuts or grooves.

Installation

- Install fabric to run with the arrow and fabric number (if applicable) on sheet side.

- Keep fabric as dry as possible during installation.
- Spread fabric out evenly across width of the machine and make certain that the guideline is straight and parallel to the CD.
- Ensure that the fabric is flat and free of wrinkles and machine debris.
- Adjust stretch roll until fabric is snug.

Start-Up

- Load press to dead weight and jog slowly while checking that the fabric is flat and free of creases and wrinkles.
- Increase fabric tension to normal running level (14–22 pli) making certain that tension does not exceed 30 pli (5.4 kilogram per cm).
- Maintain tension during start-up and check frequently.
- Wet up by applying water evenly across the fabric using a full width shower, making certain that no puddling occurs.
- When the fabric is uniformly wet, load press to normal weight and turn on vacuum equipment.

Trimming

Trimming should be delayed, if possible, in the event that the width comes in during break-in. If trimming is necessary, the following procedure is recommended:

- The fabric should be trimmed on the non-guide paddle side.
- A guide mark should be made with a pen or marker on the edge to be trimmed.
- A razor or sharp knife should be held firmly on the guide-mark where the fabric is supported
- Jog the fabric a full revolution; then cut the trimmed loop and pull it free.
- If raveling occurs after trimming, a heat-sealing iron is effective at stopping the raveling.

If a seam fabric is trimmed, special caution should be taken to ensure that the pintle is not cut so that it can be stitched back into the fabric. This will keep the seam from opening on the edges.

3.3.2 Fabric Shutdown Procedures

When a machine is scheduled for a shutdown exceeding two hours and the clothing is to remain on the machine, the following procedures will help to extend fabric life as well as prevent start-up problems.

Wash Fabric and Follow with a Thorough Rinsing

It is important to remove contaminants from the fabric. If allowed to remain in the fabric, they would harden and be difficult to remove and may affect start-up.

- Reduce the machine speed to a crawl to increase the dwell time at the suction boxes and rolls.
- Shower on the felt cleaner and allow it to act on the fabric for 15–20 minutes. Ensure that the whole fabric has been treated.
- Reduce the press load to the lowest practical level and leave the uhle box vacuum on.
- Rinse the fabric using flooding showers, lubricating showers, or high pressure showers (use low pressures so the felt is not damaged)
- Set showers at normal operating pressures for 10–15 minutes to thoroughly rinse the fabric
- Shut off the showers.
- Use vacuum boxes to dewater the felts uniformly and to the lowest practical moisture levels.
- A fabric softener or wetting agent can be applied to improve wet-up of the fabric at start-up.

Relax Fabric Tension

Once the fabric has been cleaned, rinsed, and conditioned, the showers and vacuum

should be turned off and the machine stopped. Release the fabric tension by backing off the stretch roll. This will prevent excessive tension from developing and roll deflection from occurring as the fabric dries.

Inspect Rolls, Showers and Uhle Boxes

- Inspect all roll surfaces for wear or damage which could cause abnormal wear to the fabric.
- Inspect and clean shower nozzles and uhle box covers.

3.3.3 Repairing Damaged Areas of Fabric

Occasionally during the normal operation of a paper machine, fabrics may become damaged by some object producing a hole or tear in the fabric. Due to the rising cost of downtime, it may be feasible to try and repair the damaged area temporarily until an extended downtime is scheduled. Sewing kits utilizing nylon thread or similar type yarn with curved needles may be supplied by the various paper machine clothing suppliers usually free of charge.

While there is no one method for repairing these areas, some suggestions are given below:

- The mended area should not be appreciably thicker than the fabric itself.
- In stitching, caution should be used not to pull the thread too tight causing the area to pucker, allowing for normal widening and stretching during machine operation.
- Two steps may be necessary for complete repair: closing the damaged area with stitching and then covering the closure with additional stitching.
- After stitching is completed, the ends should be left untied. A resin used for sealing edges of seam fabrics may be applied thinly to the roll side to prevent the yarns from coming loose.

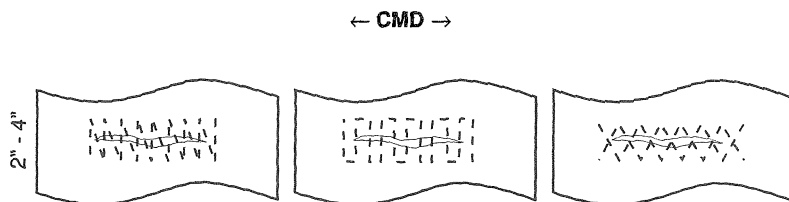


FIGURE 3.24. Damaged areas running in the cross machine direction and corresponding stitch types.

- Inspection of the damaged area as often as possible is recommended for safety purposes in case the fabric starts coming apart causing injury to personnel.

For damaged areas extending in the cross machine direction (Figure 3.24), the repair must be strong enough to prevent the fabric from ripping off the machine when run under tension. The stitch should extend vertically at least 2 inch on either side of the tear. The length will depend on the fineness of the fabric and the paper grade. Various stitching methods are shown in Figure 3.24.

If the tear is in the machine direction, the stitch will be predominantly back and forth horizontally. Since there will not be tension on the MD yarns seen in the cross machine direction shear, the stitch need to be only minimum of 1 inch on either side (Figure 3.25).

Depending on the size and direction of the tear, both vertical and horizontal stitching as a backup may be required on both types of tears. In addition, a “needle punch board” and batt can be provided for by punching the batt back and forth in the damaged area for grades such as fine paper.

3.3.4 Guiding

In general, a press fabric will line up at right angles to the surface it touches first. As shown in Figure 3.26, a guide roll angled or pivoted with the run of the press fabric will cause it to guide in the direction of the roll side that it contacts first. If the guide roll is angled in the opposite direction or pivoted away from the run of the fabric, it will cause it to track to the other side.

The above principle is valid for a normal guide roll wrap of 25° to 35° (Figure 3.27). Angles greater than recommended could cause a distance effect and skew the fabric. Angles less than recommended could result in poor fabric response to the guideline roll effect. The distance from the lead in roll to the guide roll should be approximately twice that from the guide roll to the lead-out roll. This ensures that the fabric response to the guide roll is effective.

Other Factors Influencing Guiding

Misalignment of a roll can cause guiding problems. The symptoms are typically characterized by a fabric that consistently tracks to

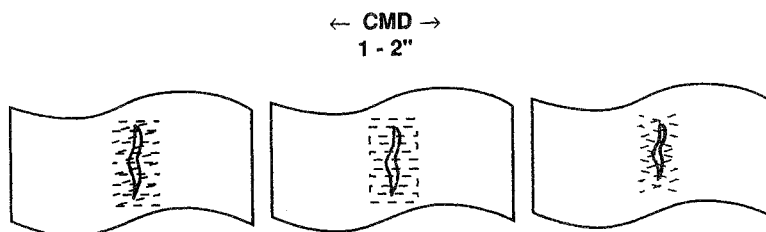


FIGURE 3.25. Machine direction tear and stitch types.

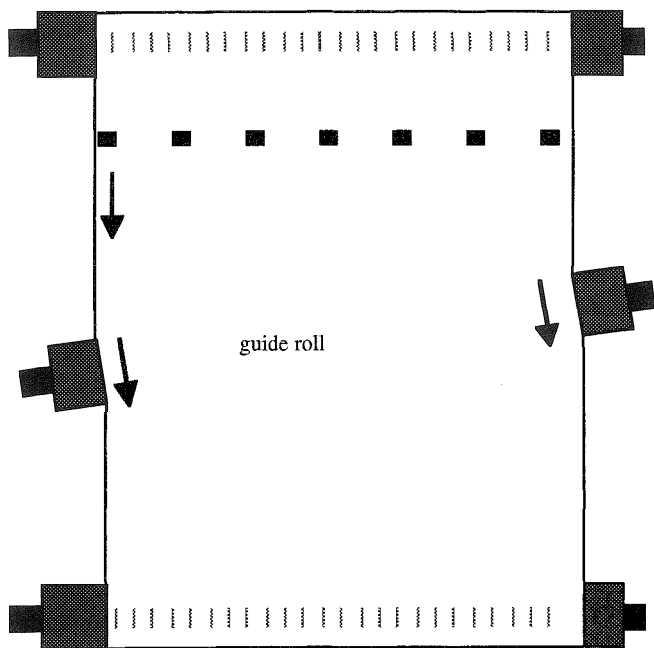


FIGURE 3.26. Press fabric guiding.

one side. It can occur with any roll but usually it is the stretch roll that is the culprit. With independent movement on each side of the stretch roll, it is inherently more likely to be skewed. The influence on the fabric is the same as by the guide roll. However, because of the large degree of wrap by the press fabric, the guideline will skew due to distance effect

that is created. The stretch roll should be checked periodically for misalignment.

Bowed Rolls

A bowed roll positioned in the same plane as the fabric run operates on the guiding principle, guiding the edges away from the center.

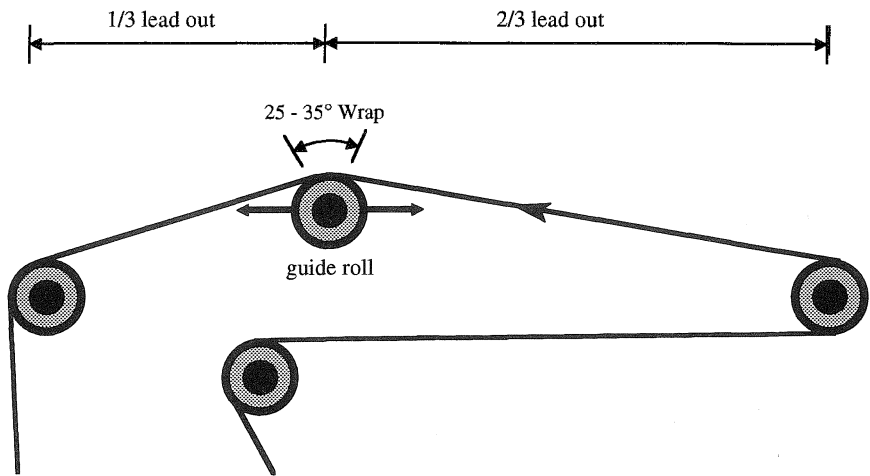


FIGURE 3.27. Lead-in and lead-out distances for guide roll.

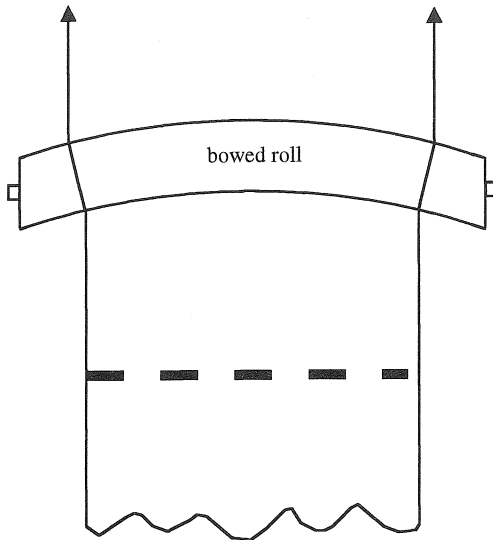


FIGURE 3.28. Widening effect of the bowed roll.

In this position, it provides the maximum widening or flattening effect on the fabric (Figure 3.28).

If the bow is turned into the plane of the fabric run, the widening and flattening effect will be lessened, proportional to the angle. A distance effect will be created, increasing proportional to the angle, causing the tradeline to lag in the center.

3.3.5 Fabric or Guideline Distortion

Press fabrics are designed to run with a straight guideline. A line that is bowed or skewed will distort the weave changing the permeability and other properties of the fabric. This could negatively impact water handling capability and result in a poorer response to conditioning and cleaning. The distortion could also cause the fabric to become unstable and possibly wrinkle.

Press fabric distortion is caused by two effects: distance-producing devices and speed-producing devices.

Distance Producing Devices

A distance-producing device is anything that causes a point on the fabric to travel a greater distance than any other point on the fabric during a revolution of the fabric. A cocked or skewed stretch roll, for instance, is such a device. However, any roll that is misaligned can cause the distance effect and fabric skew (Figure 3.29).

Side "B" would travel a shorter distance than side "A," and all points between "A" and "B" will travel progressively shorter distances per revolution than side "A." This will result in the "B" side skewing ahead. Adjustment to

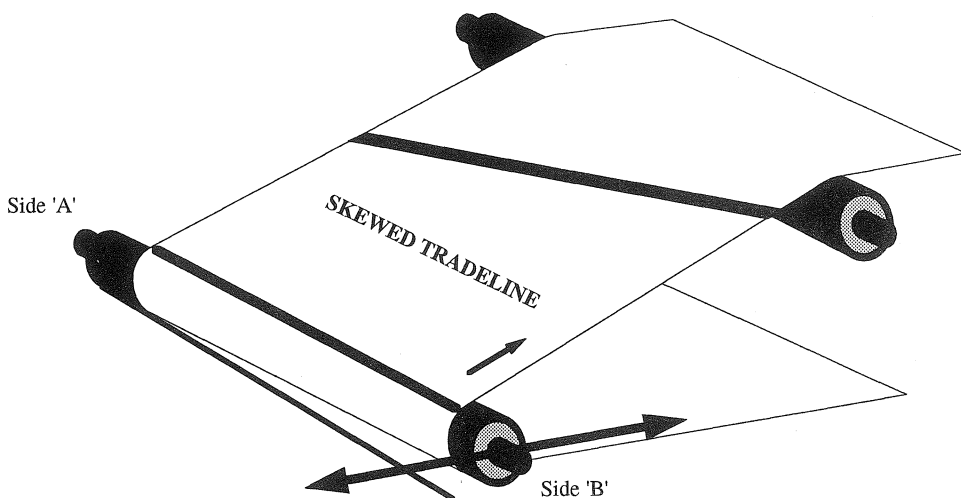


FIGURE 3.29. Schematic of a distance-producing device.

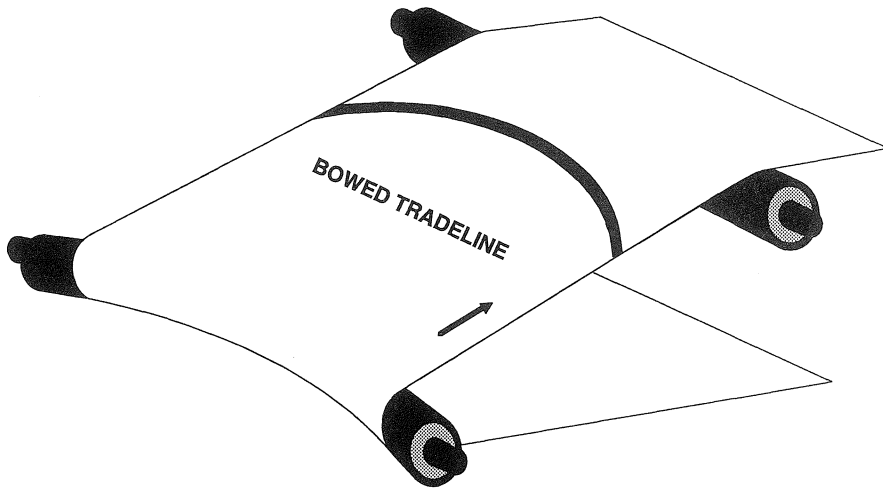


FIGURE 3.30. Roll deflection.

equal the horizontal positioning of both “A” and “B” side of the stretch roll will usually correct the skew.

Limber press fabric rolls or too high fabric tension causing roll deflection can cause the edges of the fabric to travel a greater distance than the center (Figure 3.30). The center travels a shorter distance resulting in the fabric and guideline running ahead in the center.

A bowed roll, such as Mt. Hope roll, that is turned into the press fabric run will also cause a distance effect in the center of the fabric. As the effect is greater length in the center, causing the tradeline to bow behind, the bowed roll is frequently used to correct a guide line that chronically runs ahead in the center.

Speed-Producing Devices

A speed-producing device is anything that causes a point on the fabric to travel at a different speed per revolution than any other point on the fabric. A crowned roll is such a

device (Figure 3.31). If point “B” is the center of the crowned roll and point “A” is the edge of the roll, it can be seen that point “B” must travel at a greater speed to make one revolution in the same time that it takes point “A” to make one revolution. As a result, the center and all points of greater in diameter than point “A” are pushed ahead. The result on the press fabric is a guideline that is bowed ahead. The degree of bow is proportional to the difference in circumference.

Other Speed-Producing Devices (Figure 3.32)

Concave rolls have the opposite effect as a crowned roll. The edges are greater in circumference resulting in higher surface roll speed at the edges. The effect on the fabric would be to correct a chronic problem with guideline bowing ahead.

Leadered roll is a roll that has had the surface altered to create a greater diameter. This usually involves wrapping the roll with fabric

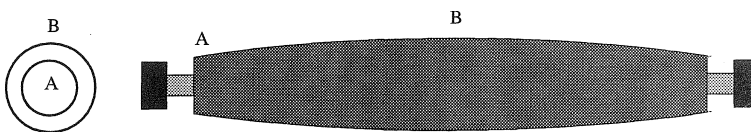


FIGURE 3.31. A crowned roll as a speed-producing device.

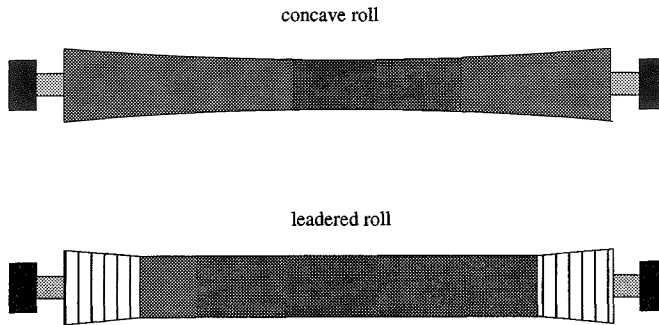


FIGURE 3.32. Concave (negative crown) and leadered rolls.

and/or tape, resembling a screw thread. Thread direction changes at the roll center line and must be set to widen, not narrow, the sheet or fabric running across it.

There may be other speed or distance producing devices that cause guideline distortion, but the point to remember is that distortion is caused by one of these devices. It follows then that one of these devices can also be used to correct distortion. Occasionally, it will be seen that fabrics from two different suppliers will show a different guideline distortion on the same position. Usually several devices on a position are affecting the fabric guideline, whose position is the net result of these several devices. The internal resistance to distortion of fabrics from different manufacturers may result in a different net effect.

3.3.6 Press Fabric Problems— Evaluation Guidelines

Table 3.5 gives a summary of common press fabric problems.

Dimensional Stability

Too Narrow

The question is whether the fabric started too narrow or became too narrow after running.

- Measure length and width (preferably at start-up and after running at least one day). If accurate measurement is not possible, determine sizes relative to fabric rolls and to amount of jack. Also note the tension (tight/slack).
- Determine if fabric was trimmed and how much.
- Measure tradeline distortion. Determine when it occurred. Is it bowed or skewed?
- Measure maximum running width; usually equal to fabric roll width.
- Measure minimum running width. This will usually be at least two inches outside maximum sheet width on either side or “wire” width for a pick-up fabric.
- Determine whether previous fabrics and succeeding fabrics exhibited the same problem.
- Determine operating temperature of press fabric or wet end.
- Determine whether steam boxes were used on press.

Too Wide

The same procedures apply for the items above, additionally:

- Check to see whether variable bowed rolls are used.
- Check to see if press fabric is worn excessively.

TABLE 3.5. Common Press Fabric Problems.

Blowing	Common on last press on machines that make lightweight fine paper and news at high speeds. The sheet rides on the fabric entering the nip. Usually during initial start-up the sheet will bubble or lift off the felt prior to entering the nip. This can cause a wrinkle in the sheet as it passes through the nip.
Drop-offs	Occurs on PU positions usually during initial start-up as the sheet rides on the bottom of the fabric from the pick-up roll to the first nip.
Bleed-through	Occurs primarily on machines that make highly refined (slow draining stock) and highly filled grades. Paper stock and/or filler material will collect on inside felt rolls causing guiding problems and vented rolls to plug.
Bounce/vibration	Typically occurs on highly loaded, hard roll presses (last press positions). The intensity of the nip makes the press sensitive to mass variations in the fabric which usually causes once or twice per revolution bounce. Most high frequency vibrations are caused by machine problems.
Shedding	Most critical on coated grades which can cause coater streaks or printing problems with the sheet of paper. It usually occurs later in life as the fabric is starting to wear and the batt fibers weakened by mechanical or chemical action.
Poor drying	A common problem on all grades in which the felt causes poor water removal in the nip, either from rewet or misapplication of design. It is usually indicated by high sheet draws and/or increased steam usage in the dryer section.
Difficult installation	Common problem on older machines in which the stiffness of the fabric makes it very difficult to bend around rolls and pack into tight spaces. Single monofilament CD structures and heavier designs make fabrics relatively stiffer.
Seam wear (mark)	Occurs with all seam products in which the batt over the seam wears prematurely, exposing the seam and base. This causes an impression mark in the sheet.
Shadow mark	Usually occurs in the first nip suction presses, where water loads are high. If flow resistance in a nip is too high, fines and filler material in the sheet will migrate to areas of low pressure causing a density variation in the sheet. It can also be caused by the suction couch roll.

Press Bounce or Vibration

Press bounce may be caused by either the press rolls or the press fabric. Improperly balanced rolls, worn rolls, corrugated rolls, or flat areas caused by leaving rolls nipped when the machine is down may cause bounce.

Answers to the following questions may reveal the cause of bounce:

- What is the machine speed? What is the frequency of bounce? (Once or twice per revolution of the press fabric? Where relative to tradelines? Once per roll revolution? Which roll?

Is it more of a high frequency chatter?)

- Was bounce present at start-up? When did it begin? Is it getting worse or better?
- Was bounce present prior to last press fabric change?
- Were any press rolls changed immediately prior to start of bouncing?
- Did the next press fabric bounce? If so, answer the first two questions again.
- Is the fabric length a multiple of a press roll circumference? (What are the roll hardnesses? What is press loading? What was the effect of cocking the tradelines?)

Water-Handling Problems

Running Wet or Speeds Down

Press water removal may be less than standard resulting in slow speeds, increased dryer steam, increased draws, crushing, or increased paper breaks. Profiles may also be poor. Causes of running wet or slow speeds can be stock-related, machine-related or clothing related. Analysis of the problem can be very complicated:

- (1) When did poor water-handling occur?
 - Were start-up speeds down? How long?
 - Were speeds slow late in life? When?
- (2) How much were speeds down?
 - What is normal speed?
 - What is normal break-in time?
- (3) Were press loads normal? If not, why?
- (4) Was water removal equipment operating normally? How much uhle box capacity was available? Measure air flows if possible. How did uhle box vacuum (inches Hg) compare to normal?
- (5) Were furnish and grades normal? Was the sheet wet on the wire?
- (6) Did problem occur after any change such as press fabric or forming fabric change?
- (7) How did the steam use compare to normal?
- (8) Did any crushing occur? In what press? Was it localized?
- (9) How was press fabric tradeline? Distorted tradelines can close up fabrics.
- (10) Is the problem localized? If so, where?
- (11) What is the effect of cutting back lube or cleaning showers?
- (12) Is nip condition flooded or dry? Is there a save-all pan? What is the condition of roll doctors or purge showers?
- (13) How did the previous and succeeding fabrics perform?

Marking

Sheet marking is typically caused by either the press rolls, press fabrics or the forming

fabric. The marks can be either a physical impression or a flow pattern.

Physical impression marks are almost always a result of nonuniform pressing by the press fabric. It can be caused by yarns in the base fabric that are too large, too widely spaced, or too rigid. If insufficient batt is used or if the fabric is worn, base fabric yarns may become exposed and mark the sheet. Physical impression marks will typically occur in the latter presses where nip intensities are usually high.

Flow pattern marks are usually a result of flow resistance in the nip that causes lateral water movement in the sheet. This causes a concentration of fines and/or filler material in the sheet that shows the vented pattern of the press roll. The area over the suction holes or the roll grooves represents zones of low pressure under load that can cause a density variation in the sheet that may be visually apparent. Flow marks usually occur in nips where water loads are high. Worn press rolls are usually the cause; however, filled fabrics or fabrics that run too wet can contribute to the problem.

Determination of the Causes of Sheet Marking

- (1) Examine sheet samples in low angle light (best at about 20°).
- (2) What is the pattern of the marks?
 - Can it be associated with the pattern of the press roll in the machine? Trace pattern from spare roll if possible or, if a grooved roll, count the number of grooves per unit length.
 - Are there large number of marks per unit length? This will distinguish a press fabric mark from a groove mark.
 - Is the direction of the mark diagonal? This will confirm or eliminate a forming fabric related mark.

- Is the mark on the top side or wire side? Which press fabrics and rolls contact which side? It needs to be determined which press fabric or roll is causing the problem.
- (3) When did the marking begin?
 - Does the date coincide with a roll change?
 - Does the date coincide with a press fabric change?
 - Have any press fabric or roll changes occurred since the beginning of the sheet mark problem?
 - Did a roll change, press fabric change, or a forming fabric change eliminate the problem?
 - (4) What are the grinding schedules of the rolls in the machine? What is the remaining roll cover thickness compared to new?
 - (5) What are the loads? Are they normal?
 - (6) What are the roll hardnesses? Are they normal?
 - (7) How does the existing caliper of the press fabric compare with the initial or edge caliper?
 - (8) Is the press fabric surface soft; is the batt worn away?
 - (9) Is the press fabric carrying more water than normal?
 - (10) In all instances obtain a sample of both the objectionable mark and the sheet that is considered satisfactory.

Fabric Wear

Excessive press fabric wear has many causes which can be divided into three categories: mechanical, chemical, and hydraulic.

- (1) Mechanical causes
 - improper roll crown/load
 - improper roll dubbing
 - bad press roll bearing
 - scissored rolls (misaligned press rolls)

- worn or rough suction box covers
 - improper high pressure shower oscillation
 - excessive shower pressure
 - abrasives in the system
 - worn or rough press rolls
 - drag on the save-all
- (2) Hydraulic causes: In some cases the press fabric is unable to handle water properly under load in which case hydraulic damage may occur to yarns and batt material.
 - (3) Chemical causes
 - reductive agents in the system
 - oxidative damage

Wrinkling

Uniform fabric tension across the entire width of the fabric is the best way to avoid press fabric wrinkling. The most common cause of uneven tension is uneven moisture distribution during press section start-ups. Proper wet up procedures will avoid slack areas or pockets that may develop wrinkles. Tradeline alignment can be an indicator of potential wrinkling. On exceptionally long draws, a supporting press fabric roll or bowed roll may be necessary.

In the event that wrinkling occurs, the press should be relieved and the fabric spread evenly across the width of the machine. Hot water or a steam hose may help flatten the wrinkle. The press should then be loaded lightly and started up slowly.

Drop-offs

Proper pick-up usually depends upon the presence of a uniform water film on the surface of the fabric to adhere the sheet to the fabric. Drop-offs can occur during the initial start-up or later in life. Many factors can influence the propensity for drop-offs:

- freeness of the stock
- press geometry
- sheet weight
- pick-up roll vacuum

Most drop-off problems occur during initial start-up of a new press fabric. In this situation the press fabric is usually designed too open or the batt component applied too coarse. Sometimes adding filler to the fabric or cutting back on uhle box vacuum will alleviate the problem. While the problem may be related to the design of the fabric, other factors could contribute to the problem:

- poor deckle trim
- improperly set pick-up roll vacuum deckle
- plugged suction pick-up roll
- improperly positioned pick-up roll
- worn forming fabric (poor sheet release)
- improperly set draws

Blowing

Blowing, like drop-offs, is influenced by the same factors and can either occur during the start-up of a new fabric or later in life. The majority of problems are usually initial and occur in the last press. Blowing is usually seen as a bubble or layer of air between the fabric and the sheet. The propensity for blowing is contributed by the press geometry. The sheet and fabric are in contact with each other entering into the nip. Air that is either entrained in the fabric or expressed backwards in the nip causes the sheet to lift off the fabric ahead of the nip. This can cause sheet wrinkles or breaks.

If the sheet blows when the fabric is new but stops after break-in, the fabric design may need to be modified to decrease the potential air volume in the fabric by increasing its density. Modifying the surface characteristics of the fabric may provide better contact between press fabric and the sheet. If the sheet does not blow when the fabric is new but blows as the fabric gets older, conditioning and cleaning efficiency may need improving or the fabric may need to be manufactured more open.

Sometimes, increasing sheet draws between the presses or running the fabric tighter can reduce or eliminate the blowing.

Crushing

Crushing may be caused by press loading too high for the flow resistance of the sheet. It also occurs when the press fabric can not accept the water at the rate it is squeezed into the fabric at the press. The result is a deformation of the sheet. On a new press fabric the problem may be too low a permeability or insufficient void volume within the press fabric. On an older fabric anything that reduces the void volume such as wear, filling or compaction contributes to crushing. Possible temporary solutions to minimize crush are to increase sheet temperature and to reduce press loads.

Troubleshooting guidelines for press section are given in Appendix C.

3.3.7 Press Fabric Conditioning

Press fabrics must be kept clean of filler materials that accumulate in their structure. Very seldom are press fabrics removed because they are worn out. They are generally removed because they are filled or compacted to the point where they no longer handle water uniformly or lack the drainage capabilities to maintain pressing efficiency. All fabrics should be continuously and mechanically conditioned from start-up to removal.

Good cleaning systems use hydraulic forces from high and low pressure showers to loosen and flush contaminants from the press fabric structure through a uhle box. A typical system includes a high pressure needle shower, a lubricating shower, and a uhle box and vacuum system (Figure 3.33). Chemical showers are sometimes used to facilitate the cleaning process.

Showers

High pressure showers provide hydraulic forces to loosen contaminants and minimize felt compaction. Continuous operation on the sheet side generally is most effective. Continuous application is recommended as it is easier to keep a fabric clean than to clean a fabric

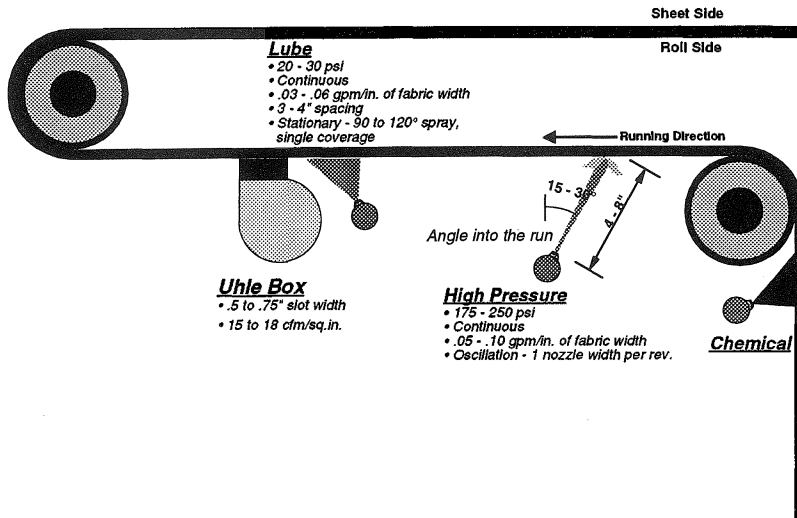


FIGURE 3.33. Press fabric conditioning system.

once it has become contaminated. Sheet side showers are more effective as the contaminants are usually filtered or trapped by the surface batt fibers and sheet side showers are more able to remove or loosen the contaminants. Roll side high pressure showers are usually not an effective means to clean the fabric. The pressures required to penetrate the base fabric and affect surface contaminants usually damage the press fabric.

Oscillator speed should be coordinated with machine speed so that the shower jet moves one nozzle width per fabric revolution. This provides complete coverage and cleaning of the fabric.

High pressure shower pumps must be interlocked with press drives so the shower will not operate when the fabric is not running, or if the oscillator fails.

Lube showers should be located immediately ahead of each uhle box on the same side of the fabric as the uhle box. These showers provide lubrication to reduce wear between the fabric and the uhle box cover.

Chemical showers provide a solvent or detergent that facilitates the cleaning process. The showers may be located either on the roll or sheet side and chemicals are applied at the point where the felt and felt roll converge.

This provides a flushing action that allows the chemical to penetrate the fabric thoroughly. Chemicals that are used continuously should be applied to the fabric as far ahead of the uhle box as practical to allow sufficient time for the cleaning solution to work.

pH of the shower water and cleaning solutions is critical in that there should not be a large differential between the stock and the showers. The pH of the sheet and showers should not vary more than ± 1 . This is to prevent pH shock that could cause precipitation of filler material. This guideline does not apply to batch cleaning as stronger solutions can be used to clean the fabric as long as it is thoroughly rinsed afterwards to neutralize pH.

Temperature of the shower water should be within $\pm 10^\circ\text{F}$ of the stock temperature. As with pH, this is to prevent upset of the sheet in the press section.

Dewatering

The typical dewatering system consists of one or more uhle boxes connected to a vacuum pump. The variables that can usually be controlled in vacuum dewatering are dwell time (slot width), pressure difference (inches

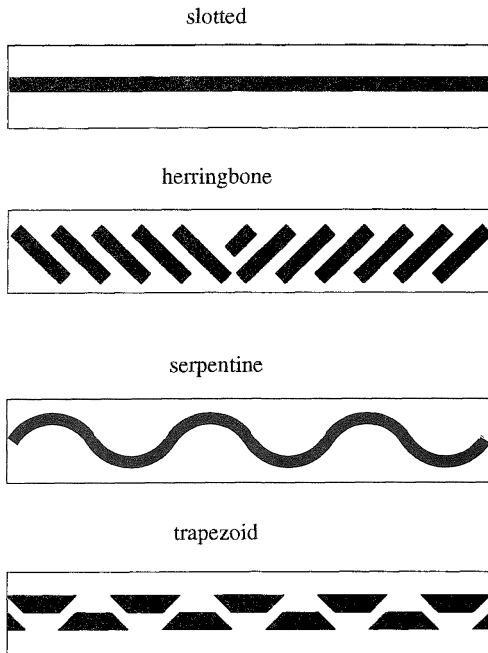


FIGURE 3.34. Common cover patterns.

of Hg) and pump capacity (cfm/sq in. of open area). Pump capacity is an important measure of dewatering energy and it should not be sacrificed to maximize one or more of the other variables. Uhle boxes do more than dewater fabrics. Showers loosen surface contaminants. The uhle box, not the showers, is the primary mechanism of fabric cleaning.

The ability to clean and dewater is related most closely to permeability. To optimize press fabric conditioning, a pump capacity of 15–18 cfm/sq in. is required. Uhle boxes on a single press fabric should be connected to a dedicated vacuum source. In nonisolated vacuum systems, air flow is proportionate to the fabric permeability.

Most uhle box covers are made from high density polyethylene. Other cover materials used are ceramic or ceramic composites. While ceramic covers are significantly more durable than polyethylene, they require greater care to prevent chipping and damage, and felt wear is usually increased. If ceramic is used, the radius of the trailing edge of the cover should be no less than 1/16 inch.

The most common cover pattern is single

or double slotted. Other patterns used are the herringbone, serpentine, and trapezoid (Figure 3.34).

The illustrated specialty patterns are used primarily on positions where seam fabrics are used. They are designed to minimize wear at the seam by providing support over the slots and reducing the degree in which the fabric is pulled into the slots. It is recommended that slot widths be no wider than 1 inch to minimize wear and shedding.

In order for a press fabric conditioning system to operate efficiently and effectively, special attention should be given to the design of the vacuum system. There are four common sources of problems with the vacuum system (Figure 3.35).

- (1) Uhle box and piping diameter: Uhle box tubes should be sized to keep the linear air velocities through the pipe below 5000 fpm. Air velocities significantly in excess of this speed induce a pressure gradient from the blanked end of the uhle box tube to the outlet side. This creates a skewed moisture profile in the fabric.
- (2) Convolution: Excessive turns and angles in the piping results in a loss of vacuum efficiency. This is important with systems that are marginal for vacuum.
- (3) Infiltration: Atmospheric air that enters the system through improperly sealed joints or holes in the piping robs the fabrics of vacuum.
- (4) Drop leg: An inch of Hg is roughly equivalent to a foot of water. If the drop leg to the seal pit lacks sufficient height, the inches of Hg at the uhle box can exceed the footage height of the drop leg. This can result in water being sucked from the seal pit back into the uhle box.

3.4 Service for Press Fabrics

3.4.1 Service Laboratories

Paper machine clothing manufacturers established customer service laboratories to enable the papermaker to learn more about the

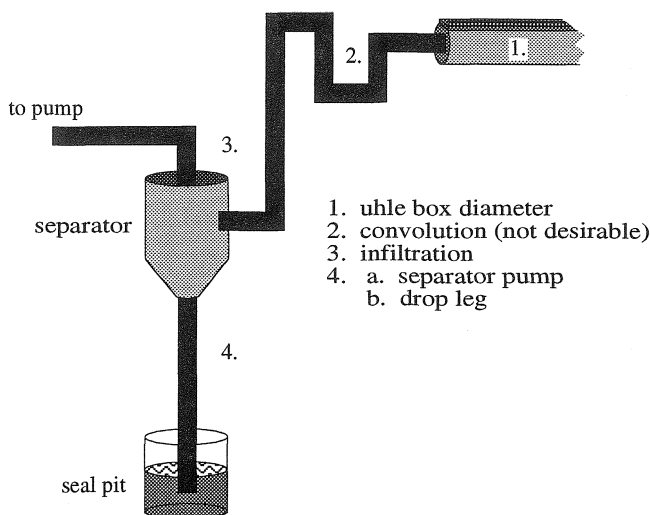


FIGURE 3.35. Typical vacuum system problem areas.

solution to press fabric problems. Working with returned, used fabrics, the laboratory often identifies the conditions that help determine the cause of the less-than-satisfactory fabric performance.

A comparison of the initial and residual measurements are made to determine the condition of the fabric. As a general rule, a laboratory analysis on a used fabric is a good tool for trend analysis and data comparison as well as to identify press section and machine-related problems.

Laboratory analysis, in general, does not provide enough information to understand application related problems. Application problems, such as initial blowing and drop-offs, that result from the design specification of the press fabric are usually diagnosed with data gathered from on machine performance compared to normal operation (Chapter 5). In some cases, the physical analysis of the fabric helps resolve application problems.

Chemical Analysis

Standard fill-up analysis determines the general type of plugging materials. These are divided into four categories:

- alkaline extractables—pitch/oil, starch, dyes, salts, dirt and detergents

- alcohol extractables—grease, oil, tar, wax, ink, pitch, and stickies (resins and latex)
- ash—alum, titanium, clay, iron, sand, talc and all inorganics
- cellulosic materials—paper stock and fines

Qualitative analysis is done for positive identification of any material in the four categories above. Infrared spectrophotography is used for specialized identification of foreign materials in the used fabric that are difficult to identify. Chemical degradation tests are done for acid, alkaline, reductive and oxidative damage.

Physical Analysis

Strength tests and profiles are done to determine the residual strength of used fabrics compared to original strength. This often reveals if the fabric could have performed longer. Profiles indicate possible problem areas such as undercrown, overcrown, misaligned rolls, worn suction boxes or any source of unusual machine wear across the width of the fabric. Weight profiles can confirm these problems as well. Caliper profiles reveal problems of uneven compaction and/or uneven physical wear. Air permeability profiles can reveal any

filled or compacted streaks across fabric width.

3.4.2 *Technical Service*

In today's practice in the papermaking industry, technical service is the primary contact a clothing supplier has with its customer and product. Outside of direct communication with the paper machine superintendent usually performed by the sales engineer, the technical service engineer is the main source of running performance information.

In general, technical service is made up of the following:

- general visual observations of the press clothing and sheet
- physical measurements of the press clothing and sheet using specialized instrumentation
- a thorough analysis of the data to translate how well the clothing is performing
- a usable report documenting findings and making pertinent recommendations

The technical service engineer reviews the press section performance and records baseline information for future comparisons. Engineers use a variety of instruments to collect baseline information. These instruments are useful to the papermaker as troubleshooting tools and as capable aids in setting and maintaining machine variables. The most widely used instruments and measurements are listed below:

- tensiometer: This instrument measures fabric tension in pli.
- running caliper gauge: The running caliper gauge is used to measure the thickness of running fabrics, which is useful when predicting fabric removal and determining loading conditions.
- gamma gauge: The gamma gauge measures the mass of the fabric. The reading obtained with the gamma gauge can be used to help determine the water balance in the press.
- water content in the fabric: Microwave measuring instruments are used to measure the amount of water in the fabric. These measurements are used to evaluate the profiles of the press fabrics.
- draw measurement gauge: Tachometer devices are used to measure clothing and sheet draw in real time to determine draw schedule throughout the machine.
- vibration analysis: Synchronous time averaging with Fast Fourier Transform (FFT) analyzer can isolate vibration sources and determine how much influence each source is having on the overall vibration in a press section.

The following TAPPI procedures can be used to analyze press and press clothing performance [7].

- (1) TIS 014-46 Press Section Monitoring: This method discusses the following parameters: fabric thickness, fabric total mass, fabric water content, sheet water to fiber ratio, degree of fabric filling/openness (permeability), running fabric length, fabric tension, fabric tradeline distortion, fabric and roll speeds, press vibration, press roll alignment, press roll crown.
- (2) TIS 014-54 Paper Machine Clothing Performance Analysis: This method provides guidelines for good clothing performance analysis

3.5 Press Types

Press types have evolved over the years to achieve faster speeds, improve water removal, and influence sheet properties. The following illustrations and discussion will comment on these press types and how they relate to the degree of two-sidedness in the sheet and how they can influence press fabric performance.

Two sidedness is defined as the percent drop in brightness from top side of the sheet to the bottom side based on the K&N ink test [8]. K&N ink test measures the percent drop

in brightness between the uninked and inked sample.

% Drop

$$= \frac{(\text{Initial Brightness} - \text{Final Brightness}) \times 100}{\text{Initial Brightness}} \quad (3.2)$$

Two Sidedness = % Drop Bottom

– % Drop Top (3.3)

The sheet from a fourdrinier is more dense on the top side due to the wash out of fines and fillers on the bottom side of the sheet. The surface density of the sheet is increased on the fabric side of each nip. The effect on sheet density is greater with each proceeding nip. Surface smoothness is used to indicate two sidedness. Sheet smoothness is measured with the Sheffield and Parker Print methods.

3.5.1 *Straight-Through™ Press* (Figure 3.36)

This press type covers an extremely wide grade and speed range as well as a range of nip conditions. Specific fabric requirements depend on water load, speed, sheet properties, sheet weights and nip conditions.

- used on virtually all grades
- variations in this type of press include inverted and reversed 2nd press (fabrics on the top side of the sheet)

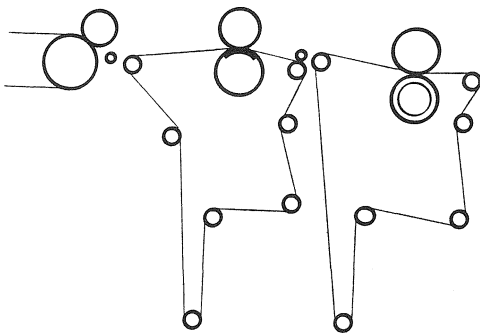


FIGURE 3.36. Straight-Through™ press.

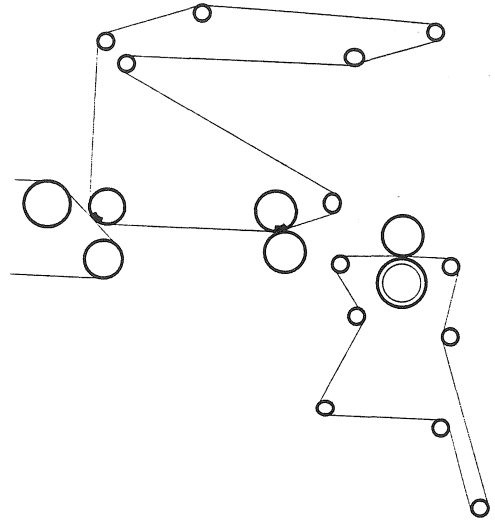


FIGURE 3.37. Inver press.

- open draws prone to sheet breaks
- prone to blowing and wad burns, if top roll not doctored
- speeds limited due to sheet breaks at open draws.
- installation a problem with multilayered fabrics

3.5.2 *Inver Press* (Figures 3.37)

This type of press is generally used to make light to medium weight fine paper grades at low to medium speeds.

- common press arrangement for fine paper grades
- prone to drop-offs with some grades
- 2nd press prone to sheet blowing
- KN: +2 to +4

3.5.3 *Twinver™ Press* (Figure 3.38)

This type press configuration was a forerunner of the no-draw cluster press. Newsprint and LWC grades at medium to high speeds are generally made on the Twinver™.

- common press arrangement for light weight, high-speed grades (news and fine)

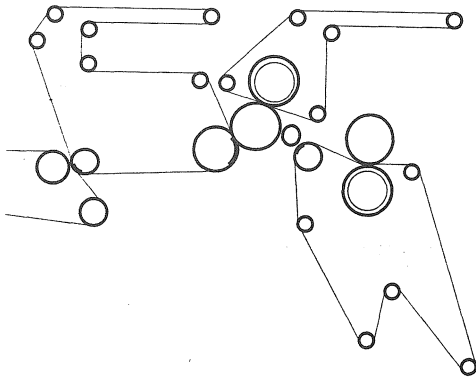


FIGURE 3.38. Twinver™ press.

- prone to sheet stealing or edge flipping by pick-up at the center roll
- 3rd press prone to vibration
- prone to drop-offs with some grades
- prone to sheet blowing in the 3rd press
- KN: +5 to +7

3.5.4 Combi Press (Figure 3.39)

This press has many of the characteristics of the Twinver™ press but without the disadvantage of a long inverted run which is prone to drop-offs. It is a compact arrangement that can be applied in a limited space.

- compact press arrangement
- no-draw effect through first two nips

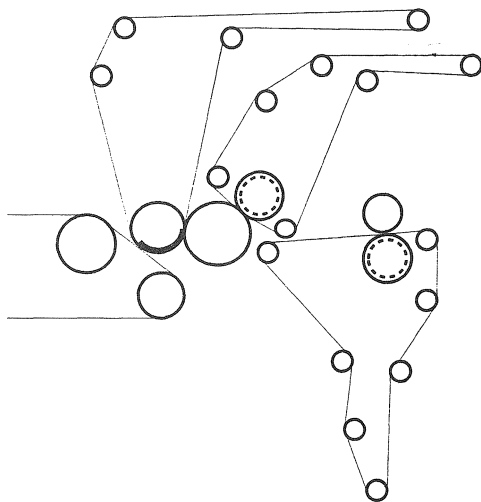


FIGURE 3.39. Combi press.

- less prone to drop-offs compared to a Twinver™ or Inver press
- pick-up prone to shadow mark with suction holes greater than 9/16 inch
- loads limited by single-felting (crush) and load differential between 1st and 2nd nip on center roll
- without 3rd press, KN: +10 to +13 (very two sided); with 3rd press KN: about +7
- difficult press fabric installations

3.5.5 Four-Roll, No-Draw Presses

TriNip™ is the most popular press arrangement for all high speed fine and newsprint grades (Figure 3.40). Double felted first nip allows for higher press loads without crush. A 4th press adds symmetry to sheet quality.

- popular no-draw, high-speed press
- double-felted first press prevents crushing and allows higher loads
- prone to shadow mark due to two nips on suction roll
- prone to sheet stealing by pick-up with some geometries
- 3rd press prone to vibration
- KN: +9 to +12 (very two-sided); the last two presses have fabrics on the top side.

TriVent™ is evolved from TriNip™ press arrangement (Figure 3.41). Separate 2nd press

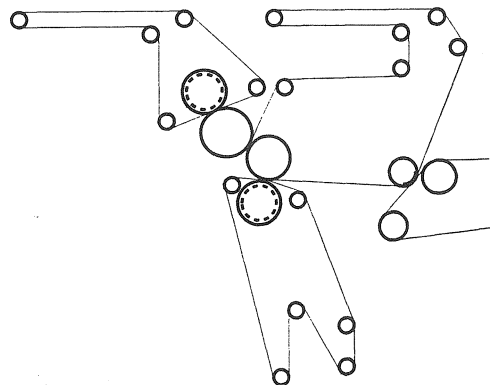


FIGURE 3.40. TriNip™.

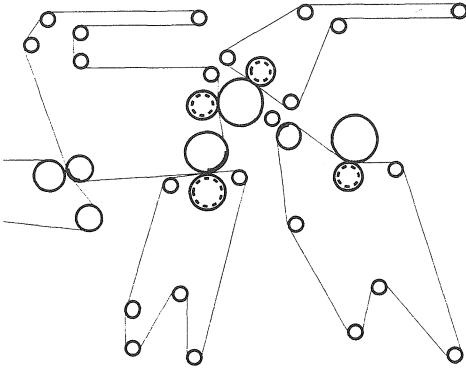


FIGURE 3.41. TriVent™.

reduces propensity for shadow mark and allows for higher loads in the 2nd and 3rd nips.

- able to obtain higher press loads in the 2nd and 3rd press nips
- prone to water rejection in the 2nd nip
- other characteristics same as TriNip™

3.5.6 Three-Roll, No Draw Presses

Alternative to TriNip™ arrangement, the BiNip™ press is popular on medium to heavy-weight grades (Figure 3.42). It improves bottom side sheet densification.

- less two-sided due to last felt on bottom side; KN: +5 to +7
- double-felted first press prevents crushing and allows higher loads
- prone to shadow mark due to two nips on suction roll

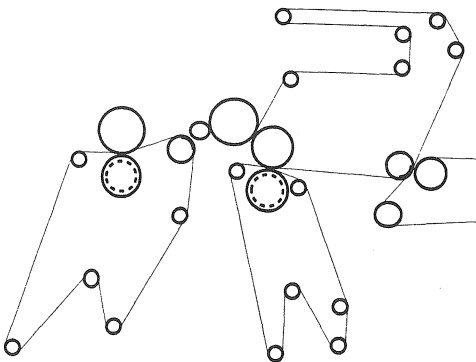


FIGURE 3.42. BiNip™.

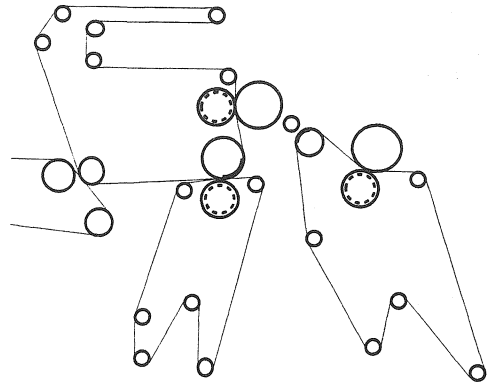


FIGURE 3.43. BiVent™.

- prone to sheet stealing by pick-up with some geometries
- 3rd press prone to blowing and wad burns

BiVent™ is evolved from the BiNip™ press arrangement (Figure 3.43). Separate 2nd press reduces propensity for shadow mark and allows for higher loads in the 2nd nip.

- prone to water rejection in the 2nd nip
- able to obtain higher press loads in the 2nd and 3rd press nips
- other characteristics same as BiNip™

3.5.7 Shoe Presses

Shoe presses were developed to increase the nip residence time for improved water removal and sheet densification. These presses operate up to 8500 pli with nip widths up to 10 inches. Sheet consistencies in some instances have improved 3 to 4% out of the press section. In all cases, sheet density is increased. The predominant grades made with shoe presses are linerboard and corrugating medium with some bleached board machines equipped with shoe presses. Recent innovations have seen application of shoe presses for fine paper and newsprint grades.

The original shoe press developed by Beloit consists of a special blanket or belt that runs inside the felt run and is pressed by a hydraulic shoe that forms the nip. Recent developments have eliminated the blanket run and contain

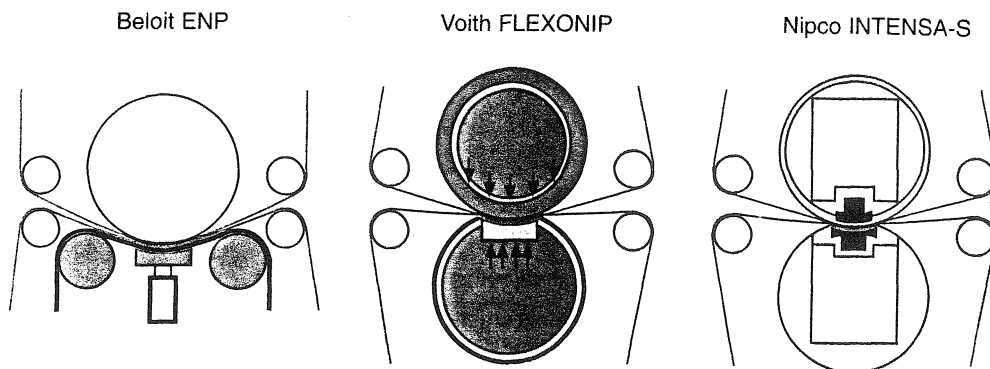


FIGURE 3.44. Shoe presses.

the shoe press inside a large diameter roll (Figure 3.44). Felt compaction and mark are potential problems with shoe presses. High water loads necessitate high void volume press fabrics.

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3.7 Review Questions

- 1 Explain how water is removed from the sheet in the press section.
- 2 What are the differences between flow controlled nips and pressure-controlled nips?
- 3 Compare the requirements of press fabrics to those of forming fabrics. Which type requires more stringent properties?
- 4 Explain the differences in manufacturing of press fabrics from forming fabrics.
- 5 What are the reasons for press fabric treatments? Explain.
- 6 Define press vibration. What are the causes of it? What are the remedies?
- 7 What are the effects of distance- and speed-producing devices?
- 8 Find out how the K&N ink test is done. What is the significance of this test?